

Chapter 5

Introduction to Matrix Algebra

diagonal matrix

"It's totally right what you say, Trix,
That in a **diagonal matrix**
You'll find," Al confirms,
"Off-diagonal terms
Are all zero. Now bug off and play, Trix!"

Bob Egg, The Omnificent English Dictionary In Limerick Form

The purpose of this chapter is to introduce you to matrix algebra, which has many applications. You are already familiar with several algebras: elementary algebra, the algebra of logic, the algebra of sets. We hope that as you studied the algebra of logic and the algebra of sets, you compared them with elementary algebra and noted that the basic laws of each are similar. We will see that matrix algebra is also similar. As in previous discussions, we begin by defining the objects in question and the basic operations.

5.1 Basic Definitions and Operations

5.1.1 Matrix Order and Equality

Definition 5.1.1 matrix. A matrix is a rectangular array of elements of the form

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{pmatrix}$$

◇

A convenient way of describing a matrix in general is to designate each entry via its position in the array. That is, the entry a_{34} is the entry in the third row and fourth column of the matrix A . Depending on the situation, we will decide in advance to which set the entries in a matrix will belong. For example, we might assume that each entry a_{ij} ($1 \leq i \leq m$, $1 \leq j \leq n$) is a real

number. In that case we would use $M_{m \times n}(\mathbb{R})$ to stand for the set of all m by n matrices whose entries are real numbers. If we decide that the entries in a matrix must come from a set S , we use $M_{m \times n}(S)$ to denote all such matrices.

Definition 5.1.2 The Order of a Matrix. A matrix A that has m rows and n columns is called an $m \times n$ (read “ m by n ”) matrix, and is said to have order $m \times n$. $\diamond$

Since it is rather cumbersome to write out the large rectangular array above each time we wish to discuss the generalized form of a matrix, it is common practice to replace the above by $A = (a_{ij})$. In general, matrices are often given names that are capital letters and the corresponding lower case letter is used for individual entries. For example the entry in the third row, second column of a matrix called C would be c_{32} .

Example 5.1.3 Orders of Some Matrices. $A = \begin{pmatrix} 2 & 3 \\ 0 & -5 \end{pmatrix}$, $B = \begin{pmatrix} 0 \\ \frac{1}{2} \\ 15 \end{pmatrix}$, and $D = \begin{pmatrix} 1 & 2 & 5 \\ 6 & -2 & 3 \\ 4 & 2 & 8 \end{pmatrix}$ are 2×2 , 3×1 , and 3×3 matrices, respectively. $\square$

Since we now understand what a matrix looks like, we are in a position to investigate the operations of matrix algebra for which users have found the most applications.

First we ask ourselves: Is the matrix $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ equal to the matrix $B = \begin{pmatrix} 1 & 2 \\ 3 & 5 \end{pmatrix}$? No, they are not because the corresponding entries in the second row, second column of the two matrices are not equal.

Next, is $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$ equal to $B = \begin{pmatrix} 1 & 2 \\ 4 & 5 \end{pmatrix}$? No, although the corresponding entries in the first two columns are identical, B doesn't have a third column to compare to that of A . We formalize these observations in the following definition.

Definition 5.1.4 Equality of Matrices. A matrix A is said to be equal to matrix B (written $A = B$) if and only if:

- (1) A and B have the same order, and
- (2) all corresponding entries are equal: that is, $a_{ij} = b_{ij}$ for all appropriate i and j .

$\diamond$

5.1.2 Matrix Addition and Scalar Multiplication

The first two operations we introduce are very natural and are not likely cause much confusion. The first is matrix addition. It seems natural that if $A = \begin{pmatrix} 1 & 0 \\ 2 & -1 \end{pmatrix}$ and $B = \begin{pmatrix} 3 & 4 \\ -5 & 2 \end{pmatrix}$, then

$$A + B = \begin{pmatrix} 1+3 & 0+4 \\ 2-5 & -1+2 \end{pmatrix} = \begin{pmatrix} 4 & 4 \\ -3 & 1 \end{pmatrix}.$$

However, if $A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \end{pmatrix}$ and $B = \begin{pmatrix} 3 & 0 \\ 2 & 8 \end{pmatrix}$, is there a natural way

to add them to give us $A + B$? No, the orders of the two matrices must be identical.

Definition 5.1.5 Matrix Addition. Let A and B be $m \times n$ matrices. Then $A + B$ is an $m \times n$ matrix where $(A + B)_{ij} = a_{ij} + b_{ij}$ (read “The i th j th entry of the matrix $A + B$ is obtained by adding the i th j th entry of A to the i th j th entry of B ”). If the orders of A and B are not identical, $A + B$ is not defined.

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In short, $A + B$ is defined if and only if A and B are of the same order.

Another frequently used operation is that of multiplying a matrix by a number, commonly called a scalar in this context. Scalars normally come from the same set as the entries in a matrix. For example, if $A \in M_{m \times n}(\mathbb{R})$, a scalar can be any real number.

Example 5.1.6 A Scalar Product. If $c = 3$ and if $A = \begin{pmatrix} 1 & -2 \\ 3 & 5 \end{pmatrix}$ and we wish to find cA , it seems natural to multiply each entry of A by 3 so that $3A = \begin{pmatrix} 3 & -6 \\ 9 & 15 \end{pmatrix}$, and this is precisely the way scalar multiplication is defined.

□

Definition 5.1.7 Scalar Multiplication. Let A be an $m \times n$ matrix and c a scalar. Then cA is the $m \times n$ matrix obtained by multiplying c times each entry of A ; that is $(cA)_{ij} = ca_{ij}$.

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5.1.3 Matrix Multiplication

A definition that is more awkward to motivate is the product of two matrices. See [Exercise 5.1.4.8](#) for an attempt to do so. In time, the reader will see that the following definition of the product of matrices will be very useful, and will provide an algebraic system that is quite similar to elementary algebra.

Definition 5.1.8 Matrix Multiplication. Let A be an $m \times n$ matrix and let B be an $n \times p$ matrix. The product of A and B , denoted by AB , is an $m \times p$ matrix whose i th row j th column entry is

$$\begin{aligned} (AB)_{ij} &= a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj} \\ &= \sum_{k=1}^n a_{ik}b_{kj} \end{aligned}$$

for $1 \leq i \leq m$ and $1 \leq j \leq p$.

◇

The mechanics of computing one entry in the product of two matrices is illustrated in [Figure 5.1.9](#).

$$\begin{array}{c}
 \text{Col. 2} \\
 \downarrow \\
 \text{Row 1} \rightarrow \begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} -6 & 2 & 4 \\ 3 & 3 & 6 \\ 1 & 4 & 5 \end{bmatrix} \\
 \\
 \text{Row 1, Col. 2 of Product} \\
 \downarrow \\
 = \begin{bmatrix} * & (1)(2) + (-1)(3) + (0)(4) & * \\ * & * & * \\ * & * & * \end{bmatrix} \\
 \\
 = \begin{bmatrix} * & -1 & * \\ * & * & * \\ * & * & * \end{bmatrix}
 \end{array}$$

Figure 5.1.9 Computation of one entry in the product of two 3 by 3 matrices

The computation of a product can take a considerable amount of time in comparison to the time required to add two matrices. Suppose that A and B are $n \times n$ matrices; then $(AB)_{ij}$ is determined performing n multiplications and $n-1$ additions. The full product takes n^3 multiplications and $n^3 - n^2$ additions. This compares with n^2 additions for the sum of two $n \times n$ matrices. The product of two 10 by 10 matrices will require 1,000 multiplications and 900 additions, clearly a job that you would assign to a computer. The sum of two matrices requires a more modest 100 additions. This analysis is based on the assumption that matrix multiplication will be done using the formula that is given in the definition. There are more advanced methods that, in theory, reduce operation counts. For example, Strassen's algorithm (en.wikipedia.org/wiki/Strassen_algorithm) computes the product of two n by n matrices in $7 \cdot 7^{\log_2 n} - 6 \cdot 4^{\log_2 n} \approx 7n^{2.808}$ operations. There are practical issues involved in actually using the algorithm in many situations. For example, round-off error can be more of a problem than with the standard formula.

Example 5.1.10 A Matrix Product. Let $A = \begin{pmatrix} 1 & 0 \\ 3 & 2 \\ -5 & 1 \end{pmatrix}$, a 3×2 matrix,

and let $B = \begin{pmatrix} 6 \\ 1 \end{pmatrix}$, a 2×1 matrix. Then AB is a 3×1 matrix:

$$AB = \begin{pmatrix} 1 & 0 \\ 3 & 2 \\ -5 & 1 \end{pmatrix} \begin{pmatrix} 6 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \cdot 6 + 0 \cdot 1 \\ 3 \cdot 6 + 2 \cdot 1 \\ -5 \cdot 6 + 1 \cdot 1 \end{pmatrix} = \begin{pmatrix} 6 \\ 20 \\ -29 \end{pmatrix}$$

□

Remarks:

- (1) The product AB is defined only if A is an $m \times n$ matrix and B is an $n \times p$ matrix; that is, the two “inner” numbers must be equal. Furthermore,

the order of the product matrix AB is the “outer” numbers, in this case $m \times p$.

- (2) It is wise to first determine the order of a product matrix. For example, if A is a 3×2 matrix and B is a 2×2 matrix, then AB is a 3×2 matrix of the form

$$AB = \begin{pmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \\ c_{31} & c_{32} \end{pmatrix}$$

Then to obtain, for example, c_{31} , we multiply corresponding entries in the third row of A times the first column of B and add the results.

Example 5.1.11 Multiplication with a diagonal matrix. Let

$$A = \begin{pmatrix} -1 & 0 \\ 0 & 3 \end{pmatrix} \text{ and } B = \begin{pmatrix} 3 & 10 \\ 2 & 1 \end{pmatrix}. \quad \text{Then } AB = \begin{pmatrix} -1 \cdot 3 + 0 \cdot 2 & -1 \cdot 10 + 0 \cdot 1 \\ 0 \cdot 3 + 3 \cdot 2 & 0 \cdot 10 + 3 \cdot 1 \end{pmatrix} = \begin{pmatrix} -3 & -10 \\ 6 & 3 \end{pmatrix}$$

The net effect is to multiply the first row of B by -1 and the second row of B by 3 .

Note: $BA = \begin{pmatrix} -3 & 30 \\ -2 & 3 \end{pmatrix} \neq AB$. The columns of B are multiplied by -1 and 3 when the order is switched. $\square$

Remarks:

- An $n \times n$ matrix is called a *square matrix*.
- If A is a square matrix, AA is defined and is denoted by A^2 , and $AAA = A^3$, etc.
- The $m \times n$ matrices whose entries are all 0 are denoted by $\mathbf{0}_{m \times n}$, or simply $\mathbf{0}$, when no confusion arises regarding the order.

5.1.4 Exercises

- Let $A = \begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 1 \\ 3 & -5 \end{pmatrix}$, and $C = \begin{pmatrix} 0 & 1 & -1 \\ 3 & -2 & 2 \end{pmatrix}$
 - Compute AB and BA .
 - Compute $A + B$ and $B + A$.
 - If $c = 3$, show that $c(A + B) = cA + cB$.
 - Show that $(AB)C = A(BC)$.
 - Compute A^2C .
 - Compute $B + \mathbf{0}$.
 - Compute $A\mathbf{0}_{2 \times 2}$ and $\mathbf{0}_{2 \times 2}A$, where $\mathbf{0}_{2 \times 2}$ is the 2×2 zero matrix.
 - Compute $0A$, where 0 is the real number (scalar) zero.
 - Let $c = 2$ and $d = 3$. Show that $(c + d)A = cA + dA$.
- Let $A = \begin{pmatrix} 1 & 0 & 2 \\ 2 & -1 & 5 \\ 3 & 2 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 2 & 3 \\ 1 & 1 & 2 \\ -1 & 3 & -2 \end{pmatrix}$, and $C =$

$$\begin{pmatrix} 2 & 1 & 2 & 3 \\ 4 & 0 & 1 & 1 \\ 3 & -1 & 4 & 1 \end{pmatrix} \text{ Compute, if possible;}$$

(a) $A - B$

(e) $CA - CB$

(b) AB

(c) $AC - BC$

(f) $C \begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix}$

(d) $A(BC)$

3. Let $A = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$. Find a matrix B such that $AB = I$ and $BA = I$, where $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.

4. Find AI and BI where I is as in Exercise 3, where $A = \begin{pmatrix} 1 & 8 \\ 9 & 5 \end{pmatrix}$ and $B = \begin{pmatrix} -2 & 3 \\ 5 & -7 \end{pmatrix}$. What do you notice?

5. Find A^3 if $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$. What is A^{15} equal to?

6.

(a) Determine I^2 and I^3 if $I = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$.

(b) What is I^n equal to for any $n \geq 1$?

(c) Prove your answer to part (b) by induction.

7.

(a) If

$$A = \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}, X = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \text{ and } B = \begin{pmatrix} 3 \\ 1 \end{pmatrix},$$

show that $AX = B$ is a way of expressing the system $\begin{matrix} 2x_1 + x_2 = 3 \\ x_1 - x_2 = 1 \end{matrix}$ using matrices.

(b) Express the following systems of equations using matrices:

(i) $\begin{matrix} 2x_1 - x_2 = 4 \\ x_1 + x_2 = 0 \end{matrix}$

(iii) $\begin{matrix} x_1 + x_2 & = & 3 \\ x_2 & = & 5 \end{matrix}$

$$x_1 + x_2 + 2x_3 = 1$$

$$x_1 + 3x_3 = 6$$

(ii) $\begin{matrix} x_1 + 2x_2 - x_3 = -1 \\ x_1 + 3x_2 + x_3 = 5 \end{matrix}$

8. In this exercise, we propose to show how matrix multiplication is a natural operation. Suppose a bakery produces bread, cakes and pies every weekday, Monday through Friday. Based on past sales history, the bakery produces various numbers of each product each day, summarized in the 5×3 matrix D . It should be noted that the order could be described as

“number of days by number of products.” For example, on Wednesday (the third day) the number of cakes (second product in our list) that are produced is $d_{3,2} = 4$.

$$D = \begin{pmatrix} 25 & 5 & 5 \\ 14 & 5 & 8 \\ 20 & 4 & 15 \\ 18 & 5 & 7 \\ 35 & 10 & 9 \end{pmatrix}$$

The main ingredients of these products are flour, sugar and eggs. We assume that other ingredients are always in ample supply, but we need to be sure to have the three main ones available. For each of the three products, The amount of each ingredient that is needed is summarized in the 3×3 , or “number of products by number of ingredients” matrix P . For example, to bake a cake (second product) we need $P_{2,1} = 1.5$ cups of flour (first ingredient). Regarding units: flour and sugar are given in cups per unit of each product, while eggs are given in individual eggs per unit of each product.

$$P = \begin{pmatrix} 2 & 0.5 & 0 \\ 1.5 & 1 & 2 \\ 1 & 1 & 1 \end{pmatrix}$$

These amounts are “made up”, so don’t used them to do your own baking!

- (a) How many cups of flour will the bakery need every Monday? Pay close attention to how you compute your answer and the units of each number.
- (b) How many eggs will the bakery need every Wednesday?
- (c) Compute the matrix product DP . What do you notice?
- (d) Suppose the costs of ingredients are \$0.12 for a cup of flour, \$0.15 for a cup of sugar and \$0.19 for one egg. How can this information be put into a matrix that can meaningfully be multiplied by one of the other matrices in this problem?

5.2 Special Types of Matrices

5.2.1 Diagonal Matrices

We have already investigated, in exercises in the previous section, one special type of matrix. That was the zero matrix, and found that it behaves in matrix algebra in an analogous fashion to the real number 0; that is, as the additive identity. We will now investigate the properties of a few other special matrices.

Definition 5.2.1 Diagonal Matrix. A square matrix D is called a diagonal matrix if $d_{ij} = 0$ whenever $i \neq j$. $\diamond$

Example 5.2.2 Some diagonal matrices. $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{pmatrix}$, $B = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -5 \end{pmatrix}$, and $I = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ are all diagonal matrices. $\square$

5.2.2 The Identity Matrix and Matrix Inverses

In the example above, the 3×3 diagonal matrix I whose diagonal entries are all 1's has the distinctive property that for any other 3×3 matrix A we have $AI = IA = A$. For example:

Example 5.2.3 Multiplying by the Identity Matrix. If $A = \begin{pmatrix} 1 & 2 & 5 \\ 6 & 7 & -2 \\ 3 & -3 & 0 \end{pmatrix}$, then $AI = \begin{pmatrix} 1 & 2 & 5 \\ 6 & 7 & -2 \\ 3 & -3 & 0 \end{pmatrix}$ and $IA = \begin{pmatrix} 1 & 2 & 5 \\ 6 & 7 & -2 \\ 3 & -3 & 0 \end{pmatrix}$. $\square$

In other words, the matrix I behaves in matrix algebra like the real number 1; that is, as a multiplicative identity. In matrix algebra, the matrix I is called simply the identity matrix. Convince yourself that if A is any $n \times n$ matrix $AI = IA = A$.

Definition 5.2.4 Identity Matrix. The $n \times n$ diagonal matrix I_n whose diagonal components are all 1's is called the identity matrix. If the context is clear, we simply use I . $\diamond$

In the set of real numbers we recall that, given a nonzero real number x , there exists a real number y such that $xy = yx = 1$. We know that real numbers commute under multiplication so that the two equations can be summarized as $xy = 1$. Further we know that $y = x^{-1} = \frac{1}{x}$. Do we have an analogous situation in $M_{n \times n}(\mathbb{R})$? Can we define the multiplicative inverse of an $n \times n$ matrix A ? It seems natural to imitate the definition of multiplicative inverse in the real numbers.

Definition 5.2.5 Matrix Inverse. Let A be an $n \times n$ matrix. If there exists an $n \times n$ matrix B such that $AB = BA = I$, then B is a multiplicative inverse of A (called simply an inverse of A) and is denoted by A^{-1} . $\diamond$

When we are doing computations involving matrices, it would be helpful to know that when we find A^{-1} , the answer we obtain is the only inverse of the given matrix. This would let us refer to *the* inverse of a matrix. We refrained from saying that in the definition, but the theorem below justifies it.

Remark: Those unfamiliar with the laws of matrix algebra should return to the following proof after they have familiarized themselves with the Laws of Matrix Algebra in Section 5.5.

Theorem 5.2.6 Inverses are unique. *The inverse of an $n \times n$ matrix A , when it exists, is unique.*

Proof. Let A be an $n \times n$ matrix. Assume to the contrary, that A has two (different) inverses, say B and C . Then

$$\begin{aligned} B &= BI && \text{Identity property of } I \\ &= B(AC) && \text{Assumption that } C \text{ is an inverse of } A \\ &= (BA)C && \text{Associativity of matrix multiplication} \\ &= IC && \text{Assumption that } B \text{ is an inverse of } A \\ &= C && \text{Identity property of } I \end{aligned}$$

Let $A = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$. What is A^{-1} ? Without too much difficulty, by trial

and error, we determine that $A^{-1} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{3} \end{pmatrix}$. This might lead us to guess that the inverse is found by taking the reciprocal of all nonzero entries of a matrix. Alas, it isn't that easy!

If $A = \begin{pmatrix} 1 & 2 \\ -3 & 5 \end{pmatrix}$, the "reciprocal rule" would tell us that the inverse of A is $B = \begin{pmatrix} 1 & \frac{1}{2} \\ \frac{-1}{3} & \frac{1}{5} \end{pmatrix}$. Try computing AB and you will see that you don't get the identity matrix. So, what *is* A^{-1} ? In order to understand more completely the notion of the inverse of a matrix, it would be beneficial to have a formula that would enable us to compute the inverse of at least a 2×2 matrix. To do this, we introduce the definition of the determinant of a 2×2 matrix.

Definition 5.2.7 Determinant of a 2 by 2 matrix. Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. The determinant of A is the number $\det A = ad - bc$. $\diamond$

In addition to $\det A$, common notation for the determinant of matrix A is $|A|$. This is particularly common when writing out the whole matrix, which case we would write $\begin{vmatrix} a & b \\ c & d \end{vmatrix}$ for the determinant of the general 2×2 matrix.

Example 5.2.8 Some determinants of two by two matrices. If $A = \begin{pmatrix} 1 & 2 \\ -3 & 5 \end{pmatrix}$ then $\det A = 1 \cdot 5 - 2 \cdot (-3) = 11$. If $B = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$ then $\det B = 1 \cdot 4 - 2 \cdot 2 = 0$. $\square$

Theorem 5.2.9 Inverse of 2 by 2 matrix. Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. If $\det A \neq 0$, then $A^{-1} = \frac{1}{\det A} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.

Proof. See Exercise 4 at the end of this section. $\blacksquare$

Example 5.2.10 Finding Inverses. Can we find the inverses of the matrices in [Example 5.2.8](#)? If $A = \begin{pmatrix} 1 & 2 \\ -3 & 5 \end{pmatrix}$ then

$$A^{-1} = \frac{1}{11} \begin{pmatrix} 5 & -2 \\ 3 & 1 \end{pmatrix} = \begin{pmatrix} \frac{5}{11} & -\frac{2}{11} \\ \frac{3}{11} & \frac{1}{11} \end{pmatrix}$$

The reader should verify that $AA^{-1} = A^{-1}A = I$.

The second matrix, B , has a determinant equal to zero. If we tried to apply the formula in [Theorem 5.2.9](#), we would be dividing by zero. For this reason, the formula can't be applied and in fact B^{-1} does not exist. $\square$

Remarks:

- In general, if A is a 2×2 matrix and if $\det A = 0$, then A^{-1} does not exist.
- A formula for the inverse of $n \times n$ matrices $n \geq 3$ can be derived that also involves $\det A$. Hence, in general, if the determinant of a matrix is zero, the matrix does not have an inverse. However the formula for even a 3×3 matrix is very long and is not the most efficient way to compute the inverse of a matrix.

- In Chapter 12 we will develop a technique to compute the inverse of a higher-order matrix, if it exists.
- Matrix inversion comes first in the hierarchy of matrix operations; therefore, AB^{-1} is $A(B^{-1})$.

5.2.3 Exercises

1. For the given matrices A find A^{-1} if it exists and verify that $AA^{-1} = A^{-1}A = I$. If A^{-1} does not exist explain why.

(a) $A = \begin{pmatrix} 1 & 3 \\ 2 & 1 \end{pmatrix}$

(b) $A = \begin{pmatrix} 6 & -3 \\ 8 & -4 \end{pmatrix}$

(c) $A = \begin{pmatrix} 1 & -3 \\ 0 & 1 \end{pmatrix}$

(d) $A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

(e) Use the definition of the inverse of a matrix to find A^{-1} : $A = \begin{pmatrix} 3 & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & -5 \end{pmatrix}$

2. For the given matrices A find A^{-1} if it exists and verify that $AA^{-1} = A^{-1}A = I$. If A^{-1} does not exist explain why.

(a) $A = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$

(b) $A = \begin{pmatrix} 0 & 1 \\ 0 & 2 \end{pmatrix}$

(c) $A = \begin{pmatrix} 1 & c \\ 0 & 1 \end{pmatrix}$

(d) $A = \begin{pmatrix} a & b \\ b & a \end{pmatrix}$, where $|a| \neq |b|$.

3.

(a) Let $A = \begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 3 & -3 \\ 2 & 1 \end{pmatrix}$. Verify that $(AB)^{-1} = B^{-1}A^{-1}$.

(b) Let A and B be $n \times n$ invertible matrices. Prove that $(AB)^{-1} = B^{-1}A^{-1}$. Why is the right side of the above statement written “backwards”? Is this necessary? Hint: Use [Theorem 5.2.6](#)

4. Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Derive the formula for A^{-1} .

5. Linearity of Determinants.

(a) Let A and B be 2-by-2 matrices. Show that $\det(AB) =$

$$(\det A)(\det B).$$

- (b) It can be shown that the statement in part (a) is true for all $n \times n$ matrices. Let A be any invertible $n \times n$ matrix. Prove that $\det(A^{-1}) = (\det A)^{-1}$. Note: The determinant of the identity matrix I_n is 1 for all n .
- (c) Verify that the equation in part (b) is true for the matrix in exercise 1(a) of this section.
6. Prove by induction that for $n \geq 1$, $\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}^n = \begin{pmatrix} a^n & 0 \\ 0 & b^n \end{pmatrix}$.
7. Use the assumptions in [Exercise 5.2.3.5](#) to prove by induction that if $n \geq 1$, $\det(A^n) = (\det A)^n$.
8. Prove: If the determinant of a matrix A is zero, then A does not have an inverse. Hint: Use the indirect method of proof and exercise 5.
- 9.
- (a) Let A, B , and D be $n \times n$ matrices. Assume that B is invertible. If $A = BDB^{-1}$, prove by induction that $A^m = BD^mB^{-1}$ is true for $m \geq 1$.
- (b) Given that $A = \begin{pmatrix} -8 & 15 \\ -6 & 11 \end{pmatrix} = B \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} B^{-1}$ where $B = \begin{pmatrix} 5 & 3 \\ 3 & 2 \end{pmatrix}$ what is A^{10} ?

5.3 Laws of Matrix Algebra

5.3.1 The Laws

The following is a summary of the basic laws of matrix operations. Assume that the indicated operations are defined; that is, that the orders of the matrices A , B and C are such that the operations make sense.

Table 5.3.1 Laws of Matrix Algebra

(1) Commutative Law of Addition	$A + B = B + A$
(2) Associative Law of Addition	$A + (B + C) = (A + B) + C$
(3) Distributive Law of a Scalar over Matrices	$c(A + B) = cA + cB$, where $c \in \mathbb{R}$.
(4) Distributive Law of Scalars over a Matrix	$(c_1 + c_2)A = c_1A + c_2A$, where $c_1, c_2 \in \mathbb{R}$.
(5) Associative Law of Scalar Multiplication	$c_1(c_2A) = (c_1 \cdot c_2)A$, where $c_1, c_2 \in \mathbb{R}$.
(6) Zero Matrix Annihilates all Products	$\mathbf{0}A = \mathbf{0}$, where $\mathbf{0}$ is the zero matrix.
(7) Zero Scalar Annihilates all Products	$0A = \mathbf{0}$, where 0 on the left is the scalar zero.
(8) Zero Matrix is an identity for Addition	$A + \mathbf{0} = A$.
(9) Negation produces additive inverses	$A + (-1)A = \mathbf{0}$.
(10) Right Distributive Law of Matrix Multiplication	$(B + C)A = BA + CA$.
(11) Left Distributive Law of Matrix Multiplication	$A(B + C) = AB + AC$.
(12) Associative Law of Multiplication	$A(BC) = (AB)C$.
(13) Identity Matrix is a Multiplicative Identity	$IA = A$ and $AI = A$.
(14) Involution Property of Inverses	If A^{-1} exists, $(A^{-1})^{-1} = A$.
(15) Inverse of Product Rule	If A^{-1} and B^{-1} exist, $(AB)^{-1} = B^{-1}A^{-1}$.

5.3.2 Commentary

Example 5.3.2 More Precise Statement of one Law. If we wished to write out each of the above laws more completely, we would specify the orders of the matrices. For example, Law 10 should read:

Let A , B , and C be $m \times n$, $n \times p$, and $n \times p$ matrices, respectively, then $A(B + C) = AB + AC$

□

Remarks:

- Notice the absence of the “law” $AB = BA$. Why?
- Is it really necessary to have both a right (No. 11) and a left (No. 10) distributive law? Why?

5.3.3 Exercises

1. Rewrite the above laws specifying as in [Example 5.3.2](#) the orders of the matrices.
2. Verify each of the Laws of Matrix Algebra using examples.
3. Let $A = \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix}$, $B = \begin{pmatrix} 3 & 7 & 6 \\ 2 & -1 & 5 \end{pmatrix}$, and $C = \begin{pmatrix} 0 & -2 & 4 \\ 7 & 1 & 1 \end{pmatrix}$. Compute the following as efficiently as possible by using any of the Laws of Matrix Algebra:
 - (a) $AB + AC$
 - (b) A^{-1}
 - (c) $A(B + C)$
 - (d) $(A^2)^{-1}$
 - (e) $(C + B)^{-1}A^{-1}$
4. Let $A = \begin{pmatrix} 7 & 4 \\ 2 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 3 & 5 \\ 2 & 4 \end{pmatrix}$. Compute the following as efficiently as possible by using any of the Laws of Matrix Algebra:
 - (a) AB
 - (b) $A + B$
 - (c) $A^2 + AB + BA + B^2$
 - (d) $B^{-1}A^{-1}$
 - (e) $A^2 + BA$
5. Let A and B be $n \times n$ matrices of real numbers. Is $A^2 - B^2 = (A - B)(A + B)$? Explain.

5.4 Matrix Oddities

5.4.1 Dissimilarities with elementary algebra

We have seen that matrix algebra is similar in many ways to elementary algebra. Indeed, if we want to solve the matrix equation $AX = B$ for the unknown X ,

we imitate the procedure used in elementary algebra for solving the equation $ax = b$. One assumption we need is that A is a square matrix that has an inverse. Notice how exactly the same properties are used in the following detailed solutions of both equations.

Table 5.4.1

Equation in the algebra of real numbers		Equation in matrix algebra
$ax = b$		$AX = B$
$a^{-1}(ax) = a^{-1}b$ if $a \neq 0$		$A^{-1}(AX) = A^{-1}B$ if A^{-1} exists
$(a^{-1}a)x = a^{-1}b$	Associative Property	$(A^{-1}A)X = A^{-1}B$
$1x = a^{-1}b$	Inverse Property	$IX = A^{-1}B$
$x = a^{-1}b$	Identity Property	$X = A^{-1}B$

Certainly the solution process for solving $AX = B$ is the same as that of solving $ax = b$.

The solution of $xa = b$ is $x = ba^{-1} = a^{-1}b$. In fact, we usually write the solution of both equations as $x = \frac{b}{a}$. In matrix algebra, the solution of $XA = B$ is $X = BA^{-1}$, which is not necessarily equal to $A^{-1}B$. So in matrix algebra, since the commutative law (under multiplication) is not true, we have to be more careful in the methods we use to solve equations.

It is clear from the above that if we wrote the solution of $AX = B$ as $X = \frac{B}{A}$, we would not know how to interpret $\frac{B}{A}$. Does it mean $A^{-1}B$ or BA^{-1} ? Because of this, A^{-1} is never written as $\frac{I}{A}$.

Observation 5.4.2 Matrix Oddities. Some of the main dissimilarities between matrix algebra and elementary algebra are that in matrix algebra:

- (1) AB may be different from BA .
- (2) There exist matrices A and B such that $AB = \mathbf{0}$, and yet $A \neq \mathbf{0}$ and $B \neq \mathbf{0}$.
- (3) There exist matrices A where $A \neq \mathbf{0}$, and yet $A^2 = \mathbf{0}$.
- (4) There exist matrices A where $A^2 = A$ with $A \neq I$ and $A \neq \mathbf{0}$
- (5) There exist matrices A where $A^2 = I$, where $A \neq I$ and $A \neq -I$

5.4.2 Exercises

1. Discuss each of the “Matrix Oddities” with respect to elementary algebra.
2. Determine 2×2 matrices which show that each of the “Matrix Oddities” are true.
3. Prove or disprove the following implications.
 - (a) $A^2 = A$ and $\det A \neq 0 \Rightarrow A = I$
 - (b) $A^2 = I$ and $\det A \neq 0 \Rightarrow A = I$ or $A = -I$.
4. Let $M_{n \times n}(\mathbb{R})$ be the set of real $n \times n$ matrices. Let $P \subseteq M_{n \times n}(\mathbb{R})$ be the subset of matrices defined by $A \in P$ if and only if $A^2 = A$. Let $Q \subseteq P$ be defined by $A \in Q$ if and only if $\det A \neq 0$.
 - (a) Determine the cardinality of Q .
 - (b) Consider the special case $n = 2$ and prove that a sufficient condition for $A \in P \subseteq M_{2 \times 2}(\mathbb{R})$ is that A has a zero determinant (i.e., A is

singular) and $\text{tr}(A) = 1$ where $\text{tr}(A) = a_{11} + a_{22}$ is the sum of the main diagonal elements of A .

(c) Is the condition of part b a necessary condition?

5. Write each of the following systems in the form $AX = B$, and then solve the systems using matrices.

(a)
$$\begin{aligned} 2x_1 + x_2 &= 3 \\ x_1 - x_2 &= 1 \end{aligned}$$

(d)
$$\begin{aligned} 2x_1 + x_2 &= 1 \\ x_1 - x_2 &= -1 \end{aligned}$$

(b)
$$\begin{aligned} 2x_1 - x_2 &= 4 \\ x_1 - x_2 &= 0 \end{aligned}$$

(e)
$$\begin{aligned} 3x_1 + 2x_2 &= 1 \\ 6x_1 + 4x_2 &= -1 \end{aligned}$$

(c)
$$\begin{aligned} 2x_1 + x_2 &= 1 \\ x_1 - x_2 &= 1 \end{aligned}$$

6. For those who know calculus:

(a) Write the series expansion for e^a centered around $a = 0$.

(b) Use the idea of exercise 6 to write what would be a plausible definition of e^A where A is an $n \times n$ matrix.

(c) If $A = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & -1 \\ 0 & 0 \end{pmatrix}$, use the series in part (b) to show that $e^A = \begin{pmatrix} e & e-1 \\ 0 & 1 \end{pmatrix}$ and $e^B = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$.

(d) Show that $e^A e^B \neq e^B e^A$.

(e) Show that $e^{A+B} = \begin{pmatrix} e & 0 \\ 0 & 1 \end{pmatrix}$.

(f) Is $e^A e^B = e^{A+B}$?

Chapter 12

More Matrix Algebra

augmented matrix

There's a Gaussian technique whose intent
Is to solve the constraints you present
As a matrix equation—
Once you've had the occasion
To write down your constants (**augment**).

Steve Ngai, The Omnificent English Dictionary In Limerick Form

In Chapter 5 we studied matrix operations and the algebra of sets and logic. We also made note of the strong resemblance of matrix algebra to elementary algebra. The reader should briefly review this material. In this chapter we shall look at a powerful matrix tool in the applied sciences, namely a technique for solving systems of linear equations. We will then use this process for determining the inverse of $n \times n$ matrices, $n \geq 2$, when they exist. We proceed with a development of the diagonalization process, with a discussion of several of its applications. Finally, we discuss the solution of linear equations over the integers modulo 2.

12.1 Systems of Linear Equations

12.1.1 Solutions

The method of solving systems of equations by matrices that we will look at is based on procedures involving equations that we are familiar with from previous mathematics courses. The main idea is to reduce a given system of equations to another simpler system that has the same solutions.

Definition 12.1.1 Solution Set. Given a system of equations involving real variables $x_1, x_2, \dots, x_n$, the solution set of the system is the set of n -tuples in $\mathbb{R}^n$, $(a_1, a_2, \dots, a_n)$ such that the substitutions $x_1 = a_1, x_2 = a_2, \dots, x_n = a_n$ make all the equations true. $\diamond$

In terms of logic, a solution set is a truth set of a system of equations, which is a proposition over n -tuples of real numbers.

In general, if the variables are from a set S , then the solution set will be a subset of S^n . For example, in number theory mathematicians study Diophantine equations, where the variables can only take on integer values instead of real values.

Definition 12.1.2 Equivalent Systems of Equations. Two systems of linear equations are called equivalent if they have the same set of solutions. $\diamond$

Example 12.1.3 Two equivalent systems. The previous definition tells us that if we know that the system

$$\begin{aligned}4x_1 + 2x_2 + x_3 &= 1 \\2x_1 + x_2 + x_3 &= 4 \\2x_1 + 2x_2 + x_3 &= 3\end{aligned}$$

is equivalent to the system

$$\begin{aligned}x_1 + 0x_2 + 0x_3 &= -1 \\0x_1 + x_2 + 0x_3 &= -1 \\0x_1 + 0x_2 + x_3 &= 7\end{aligned}$$

then both systems have the solution set $\{(-1, -1, 7)\}$. In other words, the simultaneous values $x_1 = -1$, $x_2 = -1$, and $x_3 = 7$ are the only values of the variables that make all three equations in either system true. $\square$

12.1.2 Elementary Operations on Equations

Theorem 12.1.4 Elementary Operations on Equations. *If any sequence of the following operations is performed on a system of equations, the resulting system is equivalent to the original system:*

- (a) *Interchange any two equations in the system.*
- (b) *Multiply both sides of any equation by a nonzero constant.*
- (c) *Multiply both sides of any equation by a nonzero constant and add the result to a second equation in the system, with the sum replacing the latter equation.*

Let us now use the above theorem to work out the details of [Example 12.1.3](#) and see how we can arrive at the simpler system.

The original system:

$$\begin{aligned}4x_1 + 2x_2 + x_3 &= 1 \\2x_1 + x_2 + x_3 &= 4 \\2x_1 + 2x_2 + x_3 &= 3\end{aligned}\tag{12.1.1}$$

Step 1. We will first change the coefficient of x_1 in the first equation to one and then use it as a pivot to obtain 0's for the coefficients of x_1 in Equations 2 and 3.

- Multiply Equation 1 by $\frac{1}{4}$ to obtain

$$\begin{aligned}x_1 + \frac{x_2}{2} + \frac{x_3}{4} &= \frac{1}{4} \\2x_1 + x_2 + x_3 &= 4 \\2x_1 + 2x_2 + x_3 &= 3\end{aligned}\tag{12.1.2}$$

- Multiply Equation 1 by -2 and add the result to Equation 2 to obtain

$$\begin{aligned}x_1 + \frac{x_2}{2} + \frac{x_3}{4} &= \frac{1}{4} \\0x_1 + 0x_2 + \frac{x_3}{2} &= \frac{7}{2} \\2x_1 + 2x_2 + x_3 &= 3\end{aligned}\tag{12.1.3}$$

- Multiply Equation 1 by -2 and add the result to Equation 3 to obtain

$$\begin{aligned} x_1 + \frac{x_2}{2} + \frac{x_3}{4} &= \frac{1}{4} \\ 0x_1 + 0x_2 + \frac{x_3}{2} &= \frac{7}{2} \\ 0x_1 + x_2 + \frac{x_3}{2} &= \frac{5}{2} \end{aligned} \tag{12.1.4}$$

We've explicitly written terms with zero coefficients such as $0x_1$ to make a point that all variables can be thought of as being involved in all equations. After this example is complete, we will discontinue this practice in favor of the normal practice of making these terms “disappear.”

Step 2. We would now like to proceed in a fashion analogous to Step 1; namely, multiply the coefficient of x_2 in the second equation by a suitable number so that the result is 1. Then use it as a pivot to obtain 0's as coefficients for x_2 in the first and third equations. This is clearly impossible (Why?), so we will first interchange Equations 2 and 3 and proceed as outlined above.

- Exchange Equations 2 and 3 to obtain

$$\begin{aligned} x_1 + \frac{x_2}{2} + \frac{x_3}{4} &= \frac{1}{4} \\ 0x_1 + x_2 + \frac{x_3}{2} &= \frac{5}{2} \\ 0x_1 + 0x_2 + \frac{x_3}{2} &= \frac{7}{2} \end{aligned} \tag{12.1.5}$$

- Multiply Equation 2 by $\frac{1}{2}$ and subtract the result from Equation 1 to obtain

$$\begin{aligned} x_1 + 0x_2 + 0x_3 &= -1 \\ 0x_1 + x_2 + \frac{x_3}{2} &= \frac{5}{2} \\ 0x_1 + 0x_2 + \frac{x_3}{2} &= \frac{7}{2} \end{aligned} \tag{12.1.6}$$

Step 3. Next, we will change the coefficient of x_3 in the third equation to one and then use it as a pivot to obtain 0's for the coefficients of x_3 in Equations 1 and 2. Notice that the coefficient of x_3 is already zero in Equation 1, so we have been saved some work!

- Multiply Equation 3 by 2 to obtain

$$\begin{aligned} x_1 + 0x_2 + 0x_3 &= -1 \\ 0x_1 + x_2 + \frac{x_3}{2} &= \frac{5}{2} \\ 0x_1 + 0x_2 + x_3 &= 7 \end{aligned}$$

- Multiply Equation 3 by $-1/2$ and add the result to Equation 2 to obtain

$$\begin{aligned} x_1 + 0x_2 + 0x_3 &= -1 \\ 0x_1 + x_2 + 0x_3 &= -1 \\ 0x_1 + 0x_2 + x_3 &= 7 \end{aligned} \tag{12.1.7}$$

From the system of equations at the end of Step 3, we see that the solution to the original system is $x_1 = -1$, $x_2 = -1$, and $x_3 = 7$.

12.1.3 Transition to Matrices

In the above sequence of steps, we note that the variables serve the sole purpose of keeping the coefficients in the appropriate location. This we can effect by using matrices. The matrix of the original system in our example is

$$\left(\begin{array}{ccc|c} 4 & 2 & 1 & 1 \\ 2 & 1 & 1 & 4 \\ 2 & 2 & 1 & 3 \end{array} \right)$$

where the matrix of the first three columns is called the coefficient matrix and the complete matrix is referred to as the augmented matrix. Since we are now using matrices to solve the system, we will translate [Theorem 12.1.4](#) into matrix language.

12.1.4 Elementary Row Operations

Theorem 12.1.5 Elementary Row Operations. *If any sequence of the following operations is performed on the augmented matrix of a system of equations, the resulting matrix is a system that is equivalent to the original system. The following operations on a matrix are called elementary row operations:*

- (1) *Exchange any two rows of the matrix.*
- (2) *Multiply any row of the matrix by a nonzero constant.*
- (3) *Multiply any row of the matrix by a nonzero constant and add the result to a second row, with the sum replacing that second row.*

Definition 12.1.6 Row Equivalent Matrices. Two matrices, A and B , are said to be row-equivalent if one can be obtained from the other by any sequence of zero or more elementary row operations. $\diamond$

If we use the notation R_i to stand for Row i of a matrix and $\longrightarrow$ to stand for row equivalence, then

$$A \xrightarrow{cR_i+R_j} B$$

means that the matrix B is obtained from the matrix A by multiplying the Row i of A by c and adding the result to Row j . The operation of multiplying row i by c is indicated by

$$A \xrightarrow{cR_i} B$$

while exchanging rows i and j is denoted by

$$A \xrightarrow{R_i \leftrightarrow R_j} B.$$

The matrix notation for the system given in our first example, with the

subsequent steps, is:

$$\begin{aligned}
 \left(\begin{array}{ccc|c} 4 & 2 & 1 & 1 \\ 2 & 1 & 1 & 4 \\ 2 & 2 & 1 & 3 \end{array} \right) &\xrightarrow{\frac{1}{4}R_1} \left(\begin{array}{ccc|c} 1 & \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ 2 & 1 & 1 & 4 \\ 2 & 2 & 1 & 3 \end{array} \right) \xrightarrow{-2R_1+R_2} \left(\begin{array}{ccc|c} 1 & \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ 0 & 0 & \frac{1}{2} & \frac{7}{2} \\ 2 & 2 & 1 & 3 \end{array} \right) \\
 &\xrightarrow{-2R_1+R_3} \left(\begin{array}{ccc|c} 1 & \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ 0 & 0 & \frac{1}{2} & \frac{7}{2} \\ 0 & 1 & \frac{1}{2} & \frac{5}{2} \end{array} \right) \xrightarrow{R_2 \leftrightarrow R_3} \left(\begin{array}{ccc|c} 1 & \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ 0 & 1 & \frac{1}{2} & \frac{5}{2} \\ 0 & 0 & \frac{1}{2} & \frac{7}{2} \end{array} \right) \\
 &\xrightarrow{-\frac{1}{2}R_2+R_1} \left(\begin{array}{ccc|c} 1 & 0 & 0 & -1 \\ 0 & 1 & \frac{1}{2} & \frac{5}{2} \\ 0 & 0 & \frac{1}{2} & \frac{7}{2} \end{array} \right) \xrightarrow{2R_3} \left(\begin{array}{ccc|c} 1 & 0 & 0 & -1 \\ 0 & 1 & \frac{1}{2} & \frac{5}{2} \\ 0 & 0 & 1 & 7 \end{array} \right) \\
 &\xrightarrow{-\frac{1}{2}R_3+R_2} \left(\begin{array}{ccc|c} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 7 \end{array} \right)
 \end{aligned}$$

This again gives us the solution. This procedure is called the **Gauss-Jordan elimination method**.

It is important to remember when solving any system of equations via this or any similar approach that at any step in the procedure we can rewrite the matrix in “equation format” to help us to interpret the meaning of the augmented matrix.

In our first example we found a unique solution, only one triple, namely $(-1, -1, 7)$, which satisfies all three equations. For a system involving three unknowns, are there any other possible results? To answer this question, let’s review some basic facts from analytic geometry.

The graph of a linear equation in three-dimensional space is a plane. So geometrically we can visualize the three linear equations as three planes in three-space. Certainly the three planes can intersect in a unique point, as in the first example, or two of the planes could be parallel. If two planes are parallel, there are no common points of intersection; that is, there are no triple of real numbers that will satisfy all three equations. Another possibility is that the three planes could intersect along a common axis or line. In this case, there would be an infinite number of real number triples in $\mathbb{R}^3$. Yet another possibility would be if the first two planes intersect in a line, but the line is parallel to, but not on, the third plane, giving us no solution. Finally if all three equations describe the same plane, the solution set would be that plane.

We can generalize these observations. In a system of n linear equations, n unknowns, there can be

- (1) a unique solution,
- (2) no solution, or
- (3) an infinite number of solutions.

To illustrate these points, consider the following examples:

Example 12.1.7 A system with no solutions. Find all solutions to the system

$$\begin{aligned}
 x_1 + 3x_2 + x_3 &= 2 \\
 x_1 + x_2 + 5x_3 &= 4 \\
 2x_1 + 2x_2 + 10x_3 &= 6
 \end{aligned}$$

The reader can verify that the augmented matrix of this system,

$$\left(\begin{array}{ccc|c} 1 & 3 & 1 & 2 \\ 1 & 1 & 5 & 4 \\ 2 & 2 & 10 & 6 \end{array} \right), \text{ reduces to } \left(\begin{array}{ccc|c} 1 & 3 & 1 & 2 \\ 1 & 1 & 5 & 4 \\ 0 & 0 & 0 & -2 \end{array} \right).$$

We can attempt to row-reduce this matrix further if we wish. However, any further row-reduction will not substantially change the last row, which, in equation form, is $0x_1 + 0x_2 + 0x_3 = -2$, or simply $0 = -2$. It is clear that we cannot find real numbers x_1 , x_2 , and x_3 that will satisfy this equation. Hence we cannot find real numbers that will satisfy all three original equations simultaneously. When this occurs, we say that the system has no solution, or the solution set is empty. $\square$

Example 12.1.8 A system with an infinite number of solutions. Next, let's attempt to find all of the solutions to:

$$\begin{aligned} x_1 + 6x_2 + 2x_3 &= 1 \\ 2x_1 + x_2 + 3x_3 &= 2 \\ 4x_1 + 2x_2 + 6x_3 &= 4 \end{aligned}$$

The augmented matrix for the system is

$$\left(\begin{array}{ccc|c} 1 & 6 & 2 & 1 \\ 2 & 1 & 3 & 2 \\ 4 & 2 & 6 & 4 \end{array} \right) \quad (12.1.8)$$

which reduces to

$$\left(\begin{array}{ccc|c} 1 & 0 & \frac{16}{11} & 1 \\ 0 & 1 & \frac{1}{11} & 0 \\ 0 & 0 & 0 & 0 \end{array} \right) \quad (12.1.9)$$

If we apply additional elementary row operations to this matrix, it will only become more complicated. In particular, we cannot get a one in the third row, third column. Since the matrix is in simplest form, we will express it in equation format to help us determine the solution set.

$$\begin{aligned} x_1 + \frac{16}{11}x_3 &= 1 \\ x_2 + \frac{1}{11}x_3 &= 0 \\ 0 &= 0 \end{aligned} \quad (12.1.10)$$

Any real numbers will satisfy the last equation. However, the first equation can be rewritten as $x_1 = 1 - \frac{16}{11}x_3$, which describes the coordinate x_1 in terms of x_3 . Similarly, the second equation gives x_2 in terms of x_3 . A convenient way of listing the solutions of this system is to use set notation. If we call the solution set of the system S , then

$$S = \left\{ \left(1 - \frac{16}{11}x_3, -\frac{1}{11}x_3, x_3 \right) \mid x_3 \in \mathbb{R} \right\}.$$

What this means is that if we wanted to list all solutions, we would replace x_3 by all possible numbers. Clearly, there is an infinite number of solutions, two of which are $(1, 0, 0)$ and $(-15, -1, 11)$, when x_3 takes on the values 0 and 11, respectively.

A Word Of Caution: Frequently we may can get “different-looking” answers to the same problem when a system has an infinite number of solutions. Assume the solutions set in this example is reported to be $A = \{(1 + 16x_2, x_2, -11x_3) \mid x_3 \in \mathbb{R}\}$. Certainly the result described by S looks

different from that described by A . To see whether they indeed describe the same set, we wish to determine whether every solution produced in S can be generated in A . For example, the solution generated by S when $x_3 = 11$ is $(-15, -1, 11)$. The same triple can be produced by A by taking $x_2 = -1$. We must prove that every solution described in S is described in A and, conversely, that every solution described in A is described in S . (See Exercise 6 of this section.) $\square$

To summarize the procedure in the Gauss-Jordan technique for solving systems of equations, we attempt to obtain 1's along the main diagonal of the coefficient matrix with 0's above and below the diagonal. We may find in attempting this that this objective cannot be completed, as in the last two examples we have seen. Depending on the way we interpret the results in equation form, we either recognize that no solution exists, or we identify "free variables" on which an infinite number of solutions are based. The final matrix forms that we have produced in our examples are referred to as **echelon forms**.

In practice, larger systems of linear equations are solved using computers. Generally, the Gauss-Jordan algorithm is the most useful; however, slight variations of this algorithm are also used. The different approaches share many of the same advantages and disadvantages. The two major concerns of all methods are:

- (1) minimizing inaccuracies due to round-off errors, and
- (2) minimizing computer time.

12.1.5 The Gauss-Jordan Algorithm

The accuracy of the Gauss-Jordan method can be improved by always choosing the element with the largest absolute value as the pivot element, as in the following algorithm.

Algorithm 12.1.9 The Gauss-Jordan Algorithm. *Given a matrix equation $Ax = b$, where A is $n \times m$, let C be the augmented matrix $[A|b]$. The process of row-reducing to echelon form involves performing the following algorithm where $C[i]$ is the i^{th} row of C .*

- (1) $i = 1$
- (2) $j = 1$
- (3) *while* $i \leq n$ and $j \leq m$:
 - (a) $\text{maxi} = i$
 - (b) *for* $k = i+1$ *to* n :

if $\text{abs}(C[k, j]) > \text{abs}(C[\text{maxi}, j])$: *then* $\text{maxi} = k$
 - (c) *if* $C[\text{maxi}, j] \neq 0$ *then*:
 - (i) *exchange rows* i *and* maxi
 - (ii) *divide each entry in row* i *by* $C[i, j]$
 - (iii) *for* $u = i+1$ *to* n :

subtract $C[u, j] * C[i]$ *from* $C[u]$
 - (iv) $i = i+1$
 - (d) $j = j+1$

Note 12.1.10 At the end of this algorithm, with the final form of C you can revert back to the equation form of the system and a solution should be clear. In general,

- If any row of C is all zeros, it can be ignored.
- If any row of C has all zero entries except for the entry in the $(m+1)^{\text{st}}$ position, the system has no solution. Otherwise, if a column has no pivot, the variable corresponding to it is a free variable. Variables corresponding to pivots are basic variables and can be expressed in terms of the free variables.

Example 12.1.11 If we apply [The Gauss-Jordan Algorithm](#) to the system

$$\begin{aligned} 5x_1 + x_2 + 2x_3 + x_4 &= 2 \\ 3x_1 + x_2 - 2x_3 &= 5 \\ x_1 + x_2 + 3x_3 - x_4 &= -1 \end{aligned}$$

the augmented matrix is

$$\left(\begin{array}{cccc|c} 5 & 1 & 2 & 1 & 2 \\ 3 & 1 & -2 & 0 & 5 \\ 1 & 1 & 3 & -1 & -1 \end{array} \right)$$

is reduced to

$$\left(\begin{array}{cccc|c} 1 & 0 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 1 & 0 & -\frac{3}{2} & \frac{3}{2} \\ 0 & 0 & 1 & 0 & -1 \end{array} \right)$$

Therefore, x_4 is a free variable in the solution and general solution of the system is

$$x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} - \frac{1}{2}x_4 \\ \frac{3}{2} + \frac{3}{2}x_4 \\ -1 \\ x_4 \end{pmatrix}$$

This conclusion is easy to see if you revert back to the equations that the final value the reduced matrix represents. $\square$

12.1.6 SageMath Note - Matrix Reduction

Given an augmented matrix, C , there is a matrix method called `echelon_form` that can be used to row reduce C . Here is the result for the system in [Example 12.1.11](#). In the assignment of a matrix value to C , notice that the first argument is `QQ`, which indicates that the entries should be rational numbers. As long as all the entries are rational, which is the case here since integers are rational, the row-reduced matrix will be all rational.

```
C = Matrix(QQ, [[5, 1, 2, 1, 2], [3, 1, -2, 0, 5], [1, 1, 3, -1, -1]])
C.echelon_form()
```

```
[1  ^^I0  ^^I0  ^^I1/2  ^^I1/2]
[0  ^^I1  ^^I0^^I-3/2  ^^I3/2]
[0  ^^I0  ^^I1  ^^I  0^^I-1]
```

If we don't specify the set from which entries are taken, it would assumed to be the integers and we do not get a fully row-reduced matrix. This is because the next step in working with the next output would involve multiplying row 2 by $\frac{1}{2}$ and row 3 by $\frac{1}{9}$, but these multipliers are not integers.

```
C2 = Matrix([[5, 1, 2, 1, 2], [3, 1, -2, 0, 5], [1, 1, 3, -1, -1]])
C2.echelon_form()
```

```
[ 1  1  3 -1 -1]
[ 0  2  2 -3  1]
[ 0  0  9  0 -9]
```

If we specifying real entries, the result isn't as nice and clean as the rational output.

```
C3 = Matrix(RR,[[5,1,2,1,2],[3,1,-2,0,5],[1,1,3,-1,-1]])
C3.echelon_form()
```

```
[ 1.0000000  0.0000000  0.0000000  0.5000000
 0.5000000000000000]
[ 0.0000000  1.0000000  0.0000000 -1.5000000
 1.5000000000000000]
[ 0.0000000  0.0000000  1.0000000  4.934324e-17
 -1.0000000000000000]
```

The default number of decimal places may vary from what you see here, but it can be controlled. The single small number in row three column four isn't exactly zero because of round-off but we could just set it to zero.

12.1.7 Exercises

1. Solve the following systems by describing the solution sets completely:

$$\begin{array}{ll} \text{(a)} & \begin{array}{l} 2x_1 + x_2 = 3 \\ x_1 - x_2 = 1 \end{array} & \begin{array}{l} x_1 + x_2 + 2x_3 = 1 \\ x_1 + 2x_2 - x_3 = -1 \\ x_1 + 3x_2 + x_3 = 5 \end{array} \\ & 2x_1 + x_2 + 3x_3 = 5 & \\ \text{(b)} & \begin{array}{l} 4x_1 + x_2 + 2x_3 = -1 \\ 8x_1 + 2x_2 + 4x_3 = -2 \end{array} & \begin{array}{l} \text{(d)} \quad \begin{array}{l} x_1 - x_2 + 3x_3 = 7 \\ x_1 + 3x_2 + x_3 = 4 \end{array} \end{array} \end{array}$$

2. Solve the following systems by describing the solution sets completely:

$$\begin{array}{ll} \text{(a)} & \begin{array}{l} 2x_1 + 2x_2 + 4x_3 = 2 \\ 2x_1 + x_2 + 4x_3 = 0 \\ 3x_1 + 5x_2 + x_3 = 0 \end{array} & \begin{array}{l} 6x_1 + 7x_2 + 2x_3 = 3 \\ 4x_1 + 2x_2 + x_3 = -2 \\ 6x_1 + x_2 + x_3 = 1 \end{array} \\ \text{(b)} & \begin{array}{l} 2x_1 + x_2 + 3x_3 = 2 \\ 4x_1 + x_2 + 2x_3 = -1 \\ 8x_1 + 2x_2 + 4x_3 = 4 \end{array} & \begin{array}{l} \text{(e)} \quad \begin{array}{l} x_1 + x_2 - x_3 + 2x_4 = 1 \\ x_1 + 2x_2 + 3x_3 + x_4 = 5 \\ x_1 + 3x_2 + 2x_3 - x_4 = -1 \end{array} \end{array} \\ \text{(c)} & \begin{array}{l} x_1 + x_2 + 2x_3 + x_4 = 3 \\ x_1 - x_2 + 3x_3 - x_4 = -2 \\ 3x_1 + 3x_2 + 6x_3 + 3x_4 = 9 \end{array} & \end{array}$$

3. Given the final augmented matrices below from the Gauss-Jordan Algorithm, identify the solutions sets. Identify the basic and free variables, and describe the solution set of the original system.

$$\begin{array}{ll} \text{(a)} & \left(\begin{array}{cccc|c} 1 & 0 & -5 & 0 & 1.2 \\ 0 & 1 & 4 & 0 & 2.6 \\ 0 & 0 & 0 & 1 & 4.5 \end{array} \right) & \text{(c)} & \left(\begin{array}{ccc|c} 1 & 0 & 6 & 5 \\ 0 & 1 & -2 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right) \\ \text{(b)} & \left(\begin{array}{ccc|c} 1 & 0 & 9 & 3 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 0 & 1 \end{array} \right) & \text{(d)} & \left(\begin{array}{ccc|c} 1 & 0 & 0 & -3 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & -1 \end{array} \right) \end{array}$$

- 4.
- (a) Write out the details of [Example 12.1.7](#).
 - (b) Write out the details of [Example 12.1.8](#).
 - (c) Write out the details of [Example 12.1.11](#).
5. Solve the following systems using only mod 5 arithmetic. Your solutions should be n -tuples from $\mathbb{Z}_5$.
- (a)
$$\begin{aligned} 2x_1 + x_2 &= 3 \\ x_1 + 4x_2 &= 1 \end{aligned}$$
 (compare your solution to the system in 5(a))
 - (b)
$$\begin{aligned} x_1 + x_2 + 2x_3 &= 1 \\ x_1 + 2x_2 + 4x_3 &= 4 \\ x_1 + 3x_2 + 3x_3 &= 0 \end{aligned}$$
- 6.
- (a) Use the solution set S of [Example 12.1.8](#) to list three different solutions to the given system. Then show that each of these solutions can be described by the set A in the same example.
 - (b) Prove that $S = A$.
7. Given a system of n linear equations in n unknowns in matrix form $AX = b$, prove that if b is a matrix of all zeros, then the solution set of $AX = b$ is a subgroup of $\mathbb{R}^n$.

12.2 Matrix Inversion

12.2.1 Developing the Process

In Chapter 5 we defined the inverse of an $n \times n$ matrix. We noted that not all matrices have inverses, but when the inverse of a matrix exists, it is unique. This enables us to define the inverse of an $n \times n$ matrix A as the unique matrix B such that $AB = BA = I$, where I is the $n \times n$ identity matrix. In order to get some practical experience, we developed a formula that allowed us to determine the inverse of invertible 2×2 matrices. We will now use the Gauss-Jordan procedure for solving systems of linear equations to compute the inverses, when they exist, of $n \times n$ matrices, $n \geq 2$. The following procedure for a 3×3 matrix can be generalized for $n \times n$ matrices, $n \geq 2$.

Given the matrix $A = \begin{pmatrix} 1 & 1 & 2 \\ 2 & 1 & 4 \\ 3 & 5 & 1 \end{pmatrix}$, we want to find its inverse, the matrix

$B = \begin{pmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \\ x_{31} & x_{32} & x_{33} \end{pmatrix}$, if it exists, such that $AB = I$ and $BA = I$. We will

concentrate on finding a matrix that satisfies the first equation and then verify that B also satisfies the second equation.

The equation

$$\begin{pmatrix} 1 & 1 & 2 \\ 2 & 1 & 4 \\ 3 & 5 & 1 \end{pmatrix} \begin{pmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \\ x_{31} & x_{32} & x_{33} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

is equivalent to

$$\begin{pmatrix} x_{11} + x_{21} + 2x_{31} & x_{12} + x_{22} + 2x_{32} & x_{13} + x_{23} + 2x_{33} \\ 2x_{11} + x_{21} + 4x_{31} & 2x_{12} + x_{22} + 4x_{32} & 2x_{13} + x_{23} + 4x_{33} \\ 3x_{11} + 5x_{21} + x_{31} & 3x_{12} + 5x_{22} + x_{32} & 3x_{13} + 5x_{23} + x_{33} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

By definition of equality of matrices, this gives us three systems of equations to solve. The augmented matrix of one of the systems, the one equating the first columns of the two matrices is:

$$\left(\begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 2 & 1 & 4 & 0 \\ 3 & 5 & 1 & 0 \end{array} \right) \quad (12.2.1)$$

Using the Gauss-Jordan algorithm, we have:

$$\begin{aligned} \left(\begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 2 & 1 & 4 & 0 \\ 3 & 5 & 1 & 0 \end{array} \right) &\xrightarrow{-2R_1+R_2} \left(\begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 0 & -1 & 0 & -2 \\ 3 & 5 & 1 & 0 \end{array} \right) \xrightarrow{-3R_1+R_3} \left(\begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 0 & -1 & 0 & -2 \\ 0 & 2 & -5 & -3 \end{array} \right) \\ &\xrightarrow{-1R_2} \left(\begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 2 & -5 & -3 \end{array} \right) \\ &\xrightarrow{-R_2+R_1 \text{ and } -2R_2+R_3} \left(\begin{array}{ccc|c} 1 & 0 & 2 & -1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & -5 & -7 \end{array} \right) \\ &\xrightarrow{-\frac{1}{5}R_3} \left(\begin{array}{ccc|c} 1 & 0 & 2 & -1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 7/5 \end{array} \right) \xrightarrow{-2R_3+R_1} \left(\begin{array}{ccc|c} 1 & 0 & 0 & -19/5 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 7/5 \end{array} \right) \end{aligned}$$

So $x_{11} = -19/5$, $x_{21} = 2$ and $x_{31} = 7/5$, which gives us the first column of B .

The matrix form of the system to obtain x_{12} , x_{22} , and x_{32} , the second column of B , is:

$$\left(\begin{array}{ccc|c} 1 & 1 & 2 & 0 \\ 2 & 1 & 4 & 1 \\ 3 & 5 & 1 & 0 \end{array} \right) \quad (12.2.2)$$

which reduces to

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 9/5 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -2/5 \end{array} \right) \quad (12.2.3)$$

The critical thing to note here is that the coefficient matrix in (12.2.2) is the same as the matrix in (12.2.1), hence the sequence of row operations that we used in row reduction are the same in both cases.

To determine the third column of B , we reduce

$$\left(\begin{array}{ccc|c} 1 & 1 & 2 & 0 \\ 2 & 1 & 4 & 0 \\ 3 & 5 & 1 & 1 \end{array} \right)$$

to obtain $x_{13} = 2/5$, $x_{23} = 0$ and $x_{33} = -1/5$. Here again it is important to note that the sequence of row operations used to solve this system is exactly

the same as those we used in the first system. Why not save ourselves a considerable amount of time and effort and solve all three systems simultaneously? This we can do this by augmenting the coefficient matrix by the identity matrix I . We then have, by applying the same sequence of row operations as above,

$$\left(\begin{array}{ccc|ccc} 1 & 1 & 2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 0 & 1 & 0 \\ 3 & 5 & 1 & 0 & 0 & 1 \end{array} \right) \rightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{19}{5} & \frac{9}{5} & \frac{2}{5} \\ 0 & 1 & 0 & 2 & -1 & 0 \\ 0 & 0 & 1 & \frac{7}{5} & -\frac{2}{5} & -\frac{1}{5} \end{array} \right)$$

So that

$$B = \begin{pmatrix} -\frac{19}{5} & \frac{9}{5} & \frac{2}{5} \\ 2 & -1 & 0 \\ \frac{7}{5} & -\frac{2}{5} & -\frac{1}{5} \end{pmatrix}$$

The reader should verify that $BA = I$ so that $A^{-1} = B$.

12.2.2 The General Method for Computing Inverses

As the following theorem indicates, the verification that $BA = I$ is not necessary. The proof of the theorem is beyond the scope of this text. The interested reader can find it in most linear algebra texts.

Theorem 12.2.1 *Let A be an $n \times n$ matrix. If a matrix B can be found such that $AB = I$, then $BA = I$, so that $B = A^{-1}$. In fact, to find A^{-1} , we need only find a matrix B that satisfies one of the two conditions $AB = I$ or $BA = I$.*

It is clear from Chapter 5 and our discussions in this chapter that not all $n \times n$ matrices have inverses. How do we determine whether a matrix has an inverse using this method? The answer is quite simple: the technique we developed to compute inverses is a matrix approach to solving several systems of equations simultaneously.

Example 12.2.2 Recognition of a non-invertible matrix. The reader

can verify that if $A = \begin{pmatrix} 1 & 2 & 1 \\ -1 & -2 & -1 \\ 0 & 5 & 8 \end{pmatrix}$ then the augmented matrix

$\left(\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ -1 & -2 & -1 & 0 & 1 & 0 \\ 0 & 5 & 8 & 0 & 0 & 1 \end{array} \right)$ reduces to

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 5 & 8 & 0 & 0 & 1 \end{array} \right) \quad (12.2.4)$$

Although this matrix can be row-reduced further, it is not necessary to do so since, in equation form, we have:

Table 12.2.3

$x_{11} + 2x_{21} + x_{31} = 1$	$x_{12} + 2x_{22} + x_{32} = 0$	$x_{13} + 2x_{23} + x_{33} = 0$
$0 = 1$	$0 = 1$	$0 = 0$
$5x_{21} + 8x_{31} = 0$	$5x_{22} + 8x_{32} = 0$	$5x_{23} + 8x_{33} = 1$

Clearly, there are no solutions to the first two systems, therefore A^{-1} does not exist. From this discussion it should be obvious to the reader that the zero row of the coefficient matrix together with the nonzero entry in the fourth column of that row in matrix (12.2.4) tells us that A^{-1} does not exist. $\square$

12.2.3 Exercises

1. In order to develop an understanding of the technique of this section, work out all the details of [Example 12.2.2](#).
2. Use the method of this section to find the inverses of the following matrices whenever possible. If an inverse does not exist, explain why.

$$(a) \begin{pmatrix} 1 & 2 \\ -1 & 3 \end{pmatrix}$$

$$(d) \begin{pmatrix} 1 & 2 & -1 \\ -2 & -3 & 1 \\ 1 & 4 & -3 \end{pmatrix}$$

$$(b) \begin{pmatrix} 0 & 3 & 2 & 5 \\ 1 & -1 & 4 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 3 & -1 \end{pmatrix}$$

$$(e) \begin{pmatrix} 6 & 7 & 2 \\ 4 & 2 & 1 \\ 6 & 1 & 1 \end{pmatrix}$$

$$(c) \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}$$

$$(f) \begin{pmatrix} 2 & 1 & 3 \\ 4 & 2 & 1 \\ 8 & 2 & 4 \end{pmatrix}$$

3. Use the method of this section to find the inverses of the following matrices whenever possible. If an inverse does not exist, explain why.

$$(a) \begin{pmatrix} \frac{1}{3} & 2 \\ \frac{1}{5} & -1 \end{pmatrix}$$

$$(b) \begin{pmatrix} 1 & 0 & 0 & 3 \\ 2 & -1 & 0 & 6 \\ 0 & 2 & 1 & 0 \\ 0 & -1 & 3 & 2 \end{pmatrix}$$

$$(c) \begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix}$$

$$(d) \begin{pmatrix} 1 & 0 & 0 \\ 2 & 2 & -1 \\ 1 & -1 & 1 \end{pmatrix}$$

$$(e) \begin{pmatrix} 2 & 3 & 4 \\ 3 & 4 & 5 \\ 4 & 5 & 6 \end{pmatrix}$$

$$(f) \begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{3} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \end{pmatrix}$$

4.

- (a) Find the inverses of the following matrices.

$$(i) \begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{pmatrix}$$

$$(ii) \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & \frac{5}{2} & 0 & 0 \\ 0 & 0 & \frac{1}{7} & 0 \\ 0 & 0 & 0 & \frac{3}{4} \end{pmatrix}$$

- (b) If D is a diagonal matrix whose diagonal entries are nonzero, what is D^{-1} ?

5. Express each system of equations in [Exercise 12.1.7.1](#) in the form $Ax = B$. When possible, solve each system by first finding the inverse of the matrix of coefficients.

12.3 An Introduction to Vector Spaces

12.3.1 Motivation for the study of vector spaces

When we encountered various types of matrices in Chapter 5, it became apparent that a particular kind of matrix, the diagonal matrix, was much easier to use in computations. For example, if $A = \begin{pmatrix} 2 & 1 \\ 2 & 3 \end{pmatrix}$, then A^5 can be found, but its computation is tedious. If $D = \begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix}$ then

$$D^5 = \begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix}^5 = \begin{pmatrix} 1^5 & 0 \\ 0 & 4^5 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1024 \end{pmatrix}$$

Even when presented with a non-diagonal matrix, we will see that it is sometimes possible to do a bit of work to be able to work with a diagonal matrix. This process is called **diagonalization**.

In a variety of applications it is beneficial to be able to diagonalize a matrix. In this section we will investigate what this means and consider a few applications. In order to understand when the diagonalization process can be performed, it is necessary to develop several of the underlying concepts of linear algebra.

12.3.2 Vector Spaces

By now, you realize that mathematicians tend to generalize. Once we have found a “good thing,” something that is useful, we apply it to as many different concepts as possible. In doing so, we frequently find that the “different concepts” are not really different but only look different. Four sentences in four different languages might look dissimilar, but when they are translated into a common language, they might very well express the exact same idea.

Early in the development of mathematics, the concept of a vector led to a variety of applications in physics and engineering. We can certainly picture vectors, or “arrows,” in the xy – plane and even in the three-dimensional space. Does it make sense to talk about vectors in four-dimensional space, in ten-dimensional space, or in any other mathematical situation? If so, what is the essence of a vector? Is it its shape or the rules it follows? The shape in two- or three-space is just a picture, or geometric interpretation, of a vector. The essence is the rules, or properties, we wish vectors to follow so we can manipulate them algebraically. What follows is a definition of what is called a vector space. It is a list of all the essential properties of vectors, and it is the basic definition of the branch of mathematics called linear algebra.

Definition 12.3.1 Vector Space. Let V be any nonempty set of objects. Define on V an operation, called addition, for any two elements $\mathbf{x}, \mathbf{y} \in V$, and denote this operation by $\mathbf{x} + \mathbf{y}$. Let scalar multiplication be defined for a real number $a \in \mathbb{R}$ and any element $\mathbf{x} \in V$ and denote this operation by $a\mathbf{x}$. The set V together with operations of addition and scalar multiplication is called a vector space over $\mathbb{R}$ if the following hold for all $\mathbf{x}, \mathbf{y}, \mathbf{z} \in V$, and $a, b \in \mathbb{R}$:

- $\mathbf{x} + \mathbf{y} = \mathbf{y} + \mathbf{x}$

- $(\mathbf{x} + \mathbf{y}) + \mathbf{z} = \mathbf{x} + (\mathbf{y} + \mathbf{z})$
- There exists a vector $\mathbf{0} \in V$, such that $\mathbf{x} + \mathbf{0} = \mathbf{x}$ for all $\mathbf{x} \in V$.
- For each vector $\mathbf{x} \in V$, there exists a unique vector $-\mathbf{x} \in V$, such that $-\mathbf{x} + \mathbf{x} = \mathbf{0}$.

These are the main properties associated with the operation of addition. They can be summarized by saying that $[V; +]$ is an abelian group.

The next four properties are associated with the operation of scalar multiplication and how it relates to vector addition.

- $a(\mathbf{x} + \mathbf{y}) = a\mathbf{x} + a\mathbf{y}$
- $(a + b)\mathbf{x} = a\mathbf{x} + b\mathbf{x}$
- $a(b\mathbf{x}) = (ab)\mathbf{x}$
- $1\mathbf{x} = \mathbf{x}$.

◇

In a vector space it is common to call the elements of V vectors and those from $\mathbb{R}$ scalars. Vector spaces over the real numbers are also called real vector spaces.

Example 12.3.2 A Vector Space of Matrices. Let $V = M_{2 \times 3}(\mathbb{R})$ and let the operations of addition and scalar multiplication be the usual operations of addition and scalar multiplication on matrices. Then V together with these operations is a real vector space. The reader is strongly encouraged to verify the definition for this example before proceeding further (see Exercise 3 of this section). Note we can call the elements of $M_{2 \times 3}(\mathbb{R})$ vectors even though they are not arrows. □

Example 12.3.3 The Vector Space $\mathbb{R}^2$. Let $\mathbb{R}^2 = \{(a_1, a_2) \mid a_1, a_2 \in \mathbb{R}\}$. If we define addition and scalar multiplication the natural way, that is, as we would on 1×2 matrices, then $\mathbb{R}^2$ is a vector space over $\mathbb{R}$. See [Exercise 12.3.3.4](#) of this section.

In this example, we have the “bonus” that we can illustrate the algebraic concept geometrically. In mathematics, a “geometric bonus” does not always occur and is not necessary for the development or application of the concept. However, geometric illustrations are quite useful in helping us understand concepts and should be utilized whenever available.

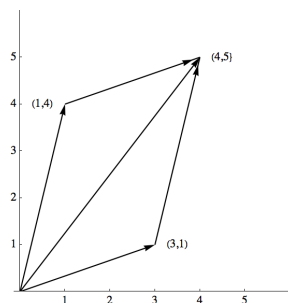


Figure 12.3.4 Sum of two vectors in $\mathbb{R}^2$

Let's consider some illustrations of the vector space $\mathbb{R}^2$. Let $\mathbf{x} = (1, 4)$ and $\mathbf{y} = (3, 1)$. We illustrate the vector (a_1, a_2) as a directed line segment, or “arrow,” from the point $(0, 0)$ to the point (a_1, a_2) . The vectors $\mathbf{x}$ and $\mathbf{y}$ are as shown in [Figure 12.3.4](#) together with $\mathbf{x} + \mathbf{y} = (1, 4) + (3, 1) = (4, 5)$. The vector

$2\mathbf{x} = 2(1, 4) = (2, 8)$ is a vector in the same direction as $\mathbf{x}$, but with twice its length. $\square$

Note 12.3.5

- (1) The common convention is to use that boldface letters toward the end of the alphabet for vectors, while letters early in the alphabet are scalars.
- (2) A common alternate notation for vectors is to place an arrow about a variable to indicate that it is a vector such as this: $\vec{x}$.
- (3) The vector $(a_1, a_2, \dots, a_n) \in \mathbb{R}^n$ is referred to as an n -tuple.
- (4) For those familiar with vector calculus, we are expressing the vector $x = a_1\hat{\mathbf{i}} + a_2\hat{\mathbf{j}} + a_3\hat{\mathbf{k}} \in \mathbb{R}^3$ as (a_1, a_2, a_3) . This allows us to discuss vectors in $\mathbb{R}^n$ in much simpler notation.

In many situations a vector space V is given and we would like to describe the whole vector space by the smallest number of essential reference vectors. An example of this is the description of $\mathbb{R}^2$, the xy -plane, via the x and y axes. Again our concepts must be algebraic in nature so we are not restricted solely to geometric considerations.

Definition 12.3.6 Linear Combination. A vector $\mathbf{y}$ in vector space V (over $\mathbb{R}$) is a linear combination of the vectors $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$ if there exist scalars $a_1, a_2, \dots, a_n$ in $\mathbb{R}$ such that $\mathbf{y} = a_1\mathbf{x}_1 + a_2\mathbf{x}_2 + \dots + a_n\mathbf{x}_n$ $\diamond$

Example 12.3.7 A Basic Example. The vector $(2, 3)$ in $\mathbb{R}^2$ is a linear combination of the vectors $(1, 0)$ and $(0, 1)$ since $(2, 3) = 2(1, 0) + 3(0, 1)$. $\square$

Example 12.3.8 A little less obvious example. Prove that the vector $(4, 5)$ is a linear combination of the vectors $(3, 1)$ and $(1, 4)$.

By the definition we must show that there exist scalars a_1 and a_2 such that:

$$\begin{aligned} (4, 5) &= a_1(3, 1) + a_2(1, 4) &\Rightarrow & 3a_1 + a_2 = 4 \\ &= (3a_1 + a_2, a_1 + 4a_2) && a_1 + 4a_2 = 5 \end{aligned}$$

This system has the solution $a_1 = 1, a_2 = 1$.

Hence, if we replace a_1 and a_2 both by 1, then the two vectors $(3, 1)$ and $(1, 4)$ produce, or generate, the vector $(4, 5)$. Of course, if we replace a_1 and a_2 by different scalars, we can generate more vectors from $\mathbb{R}^2$. If, for example, $a_1 = 3$ and $a_2 = -2$, then

$$a_1(3, 1) + a_2(1, 4) = 3(3, 1) + (-2)(1, 4) = (9, 3) + (-2, -8) = (7, -5)$$

$\square$

Will the vectors $(3, 1)$ and $(1, 4)$ generate any vector we choose in $\mathbb{R}^2$? To see if this is so, we let (b_1, b_2) be an arbitrary vector in $\mathbb{R}^2$ and see if we can always find scalars a_1 and a_2 such that $a_1(3, 1) + a_2(1, 4) = (b_1, b_2)$. This is equivalent to solving the following system of equations:

$$\begin{aligned} 3a_1 + a_2 &= b_1 \\ a_1 + 4a_2 &= b_2 \end{aligned}$$

which always has solutions for a_1 and a_2 , regardless of the values of the real numbers b_1 and b_2 . Why? We formalize this situation in a definition:

Definition 12.3.9 Generation of a Vector Space. Let $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$ be a set of vectors in a vector space V over $\mathbb{R}$. This set is said to **generate**,

or span, V if, for any given vector $\mathbf{y} \in V$, we can always find scalars $a_1, a_2, \dots, a_n$ such that $\mathbf{y} = a_1\mathbf{x}_1 + a_2\mathbf{x}_2 + \dots + a_n\mathbf{x}_n$. A set that generates a vector space is called a **generating set**. $\diamond$

We now give a geometric interpretation of the previous examples.

We know that the standard coordinate system, x axis and y axis, were introduced in basic algebra in order to describe all points in the xy -plane algebraically. It is also quite clear that to describe any point in the plane we need exactly two axes.

We can set up a new coordinate system in the following way. Draw the vector $(3, 1)$ and an axis from the origin through $(3, 1)$ and label it the x' axis. Also draw the vector $(1, 4)$ and an axis from the origin through $(1, 4)$ to be labeled the y' axis. Draw the coordinate grid for the axis, that is, lines parallel, and let the unit lengths of this “new” plane be the lengths of the respective vectors, $(3, 1)$ and $(1, 4)$, so that we obtain Figure 12.3.10.

From Example 12.3.8 and Figure 12.3.10, we see that any vector on the plane can be described using the standard xy -axes or our new $x'y'$ -axes. Hence the position which had the name $(3, 1)$ in reference to the standard axes has the name $(1, 0)$ with respect to the $x'y'$ axes, or, in the phraseology of linear algebra, the coordinates of the point $(1, 4)$ with respect to the $x'y'$ axes are $(1, 0)$.

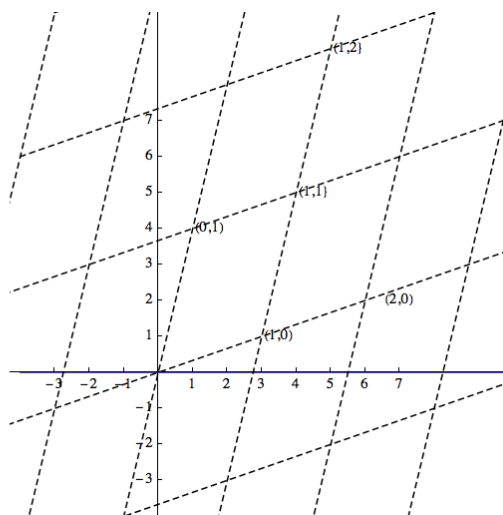


Figure 12.3.10 Two sets of axes for the plane

Example 12.3.11 One point, Two position descriptions. From Example 12.3.8 we found that if we choose $a_1 = 1$ and $a_2 = 1$, then the two vectors $(3, 1)$ and $(1, 4)$ generate the vector $(4, 5)$. Another geometric interpretation of this problem is that the coordinates of the position $(4, 5)$ with respect to the $x'y'$ axes of Figure 12.3.10 is $(1, 1)$. In other words, a position in the plane has the name $(4, 5)$ in reference to the xy -axes and the same position has the name $(1, 1)$ in reference to the $x'y'$ axes.

From the above, it is clear that we can use different axes to describe points or vectors in the plane. No matter what choice we use, we want to be able to describe each position in a unique manner. This is not the case in Figure 12.3.12. Any point in the plane could be described via the $x'y'$ axes, the $x'z'$ axes or the $y'z'$ axes. Therefore, in this case, a single point would have three different names, a very confusing situation.

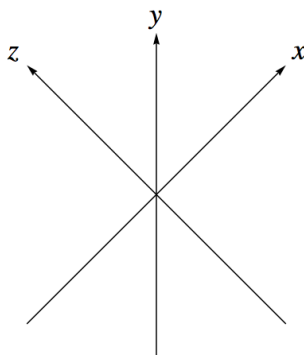


Figure 12.3.12 Three axes on a plane

□

We formalize the our observations in the previous examples in two definitions and a theorem.

Definition 12.3.13 Linear Independence/Linear Dependence. A set of vectors $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$ from a real vector space V is **linearly independent** if the only solution to the equation $a_1\mathbf{x}_1 + a_2\mathbf{x}_2 + \dots + a_n\mathbf{x}_n = \mathbf{0}$ is $a_1 = a_2 = \dots = a_n = 0$. Otherwise the set is called a **linearly dependent** set. ◇

Definition 12.3.14 Basis. A set of vectors $B = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$ is a basis for a vector space V if:

- (1) B generates V , and
- (2) B is linearly independent.

◇

Theorem 12.3.15 The fundamental property of a basis. *If $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$ is a basis for a vector space V over $\mathbb{R}$, then any vector $y \in V$ can be uniquely expressed as a linear combination of the $\mathbf{x}_i$'s.*

Proof. Assume that $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$ is a basis for V over $\mathbb{R}$. We must prove two facts:

- (1) each vector $y \in V$ can be expressed as a linear combination of the $\mathbf{x}_i$'s, and
- (2) each such expression is unique.

Part 1 is trivial since a basis, by its definition, must generate all of V .

The proof of part 2 is a bit more difficult. We follow the standard approach for any uniqueness facts. Let y be any vector in V and assume that there are two different ways of expressing y , namely

$$y = a_1\mathbf{x}_1 + a_2\mathbf{x}_2 + \dots + a_n\mathbf{x}_n$$

and

$$y = b_1\mathbf{x}_1 + b_2\mathbf{x}_2 + \dots + b_n\mathbf{x}_n$$

where at least one a_i is different from the corresponding b_i . Then equating these two linear combinations we get

$$a_1\mathbf{x}_1 + a_2\mathbf{x}_2 + \dots + a_n\mathbf{x}_n = b_1\mathbf{x}_1 + b_2\mathbf{x}_2 + \dots + b_n\mathbf{x}_n$$

so that

$$(a_1 - b_1)\mathbf{x}_1 + (a_2 - b_2)\mathbf{x}_2 + \dots + (a_n - b_n)\mathbf{x}_n = \mathbf{0}$$

Now a crucial observation: since the $\mathbf{x}'_i$ s form a linearly independent set, the only solution to the previous equation is that each of the coefficients must equal zero, so $a_i - b_i = 0$ for $i = 1, 2, \dots, n$. Hence $a_i = b_i$, for all i . This contradicts our assumption that at least one a_i is different from the corresponding b_i , so each vector $\mathbf{y} \in V$ can be expressed in one and only one way. ■

This theorem, together with the previous examples, gives us a clear insight into the significance of linear independence, namely uniqueness in representing any vector.

Example 12.3.16 Another basis for $\mathbb{R}^2$. Prove that $\{(1, 1), (-1, 1)\}$ is a basis for $\mathbb{R}^2$ over $\mathbb{R}$ and explain what this means geometrically.

First we show that the vectors $(1, 1)$ and $(-1, 1)$ generate all of $\mathbb{R}^2$. We can do this by imitating [Example 12.3.8](#) and leave it to the reader (see [Exercise 12.3.3.10](#) of this section). Secondly, we must prove that the set is linearly independent.

Let a_1 and a_2 be scalars such that $a_1(1, 1) + a_2(-1, 1) = (0, 0)$. We must prove that the only solution to the equation is that a_1 and a_2 must both equal zero. The above equation becomes $(a_1 - a_2, a_1 + a_2) = (0, 0)$ which gives us the system

$$\begin{aligned} a_1 - a_2 &= 0 \\ a_1 + a_2 &= 0 \end{aligned}$$

The augmented matrix of this system reduces in such way that the only solution is the trivial one of all zeros:

$$\left(\begin{array}{cc|c} 1 & -1 & 0 \\ 1 & 1 & 0 \end{array} \right) \longrightarrow \left(\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0 \end{array} \right) \Rightarrow a_1 = a_2 = 0$$

Therefore, the set is linearly independent. □

To explain the results geometrically, note through Exercise 12, part a, that the coordinates of each vector $\mathbf{y} \in \mathbb{R}^2$ can be determined uniquely using the vectors $(1, 1)$ and $(-1, 1)$. The concept of dimension is quite obvious for those vector spaces that have an immediate geometric interpretation. For example, the dimension of $\mathbb{R}^2$ is two and that of $\mathbb{R}^3$ is three. How can we define the concept of dimension algebraically so that the resulting definition correlates with that of $\mathbb{R}^2$ and $\mathbb{R}^3$? First we need a theorem, which we will state without proof.

Theorem 12.3.17 Basis Size is Constant. *If V is a vector space with a basis containing n elements, then all bases of V contain n elements.*

Definition 12.3.18 Dimension of a Vector Space. Let V be a vector space over $\mathbb{R}$ with basis $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$. Then the dimension of V is n . We use the notation $\dim V = n$ to indicate that V is n -dimensional. ◇

12.3.3 Exercises

1. If $a = 2$, $b = -3$, $A = \begin{pmatrix} 1 & 0 & -1 \\ 2 & 3 & 4 \end{pmatrix}$, $B = \begin{pmatrix} 2 & -2 & 3 \\ 4 & 5 & 8 \end{pmatrix}$, and $C = \begin{pmatrix} 1 & 0 & 0 \\ 3 & 2 & -2 \end{pmatrix}$ verify that all properties of the definition of a vector space are true for $M_{2 \times 3}(\mathbb{R})$ with these values.
2. Let $a = 3$, $b = 4$, $\mathbf{x} = (-1, 3)$, $\mathbf{y} = (2, 3)$, and $\mathbf{z} = (1, 0)$. Verify that all properties of the definition of a vector space are true for $\mathbb{R}^2$ for these values.

3.

- (a) Verify that $M_{2 \times 3}(\mathbb{R})$ is a vector space over $\mathbb{R}$. What is its dimension?
- (b) Is $M_{m \times n}(\mathbb{R})$ a vector space over $\mathbb{R}$? If so, what is its dimension?

4.

- (a) Verify that $\mathbb{R}^2$ is a vector space over $\mathbb{R}$.
- (b) Is $\mathbb{R}^n$ a vector space over $\mathbb{R}$ for every positive integer n ?

5. Let $P^3 = \{a_0 + a_1x + a_2x^2 + a_3x^3 \mid a_0, a_1, a_2, a_3 \in \mathbb{R}\}$; that is, P^3 is the set of all polynomials in x having real coefficients with degree less than or equal to three. Verify that P^3 is a vector space over $\mathbb{R}$. What is its dimension?

6. For each of the following, express the vector $\mathbf{y}$ as a linear combination of the vectors $\mathbf{x}_1$ and $\mathbf{x}_2$.

- (a) $\mathbf{y} = (5, 6)$, $\mathbf{x}_1 = (1, 0)$, and $\mathbf{x}_2 = (0, 1)$
- (b) $\mathbf{y} = (2, 1)$, $\mathbf{x}_1 = (2, 1)$, and $\mathbf{x}_2 = (1, 1)$
- (c) $\mathbf{y} = (3, 4)$, $\mathbf{x}_1 = (1, 1)$, and $\mathbf{x}_2 = (-1, 1)$

7. Express the vector $\begin{pmatrix} 1 & 2 \\ -3 & 3 \end{pmatrix} \in M_{2 \times 2}(\mathbb{R})$, as a linear combination of $\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$, $\begin{pmatrix} -1 & 5 \\ 2 & 1 \end{pmatrix}$, $\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$ and $\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$

8. Express the vector $x^3 - 4x^2 + 3 \in P^3$ as a linear combination of the vectors $1, x, x^2$, and x^3 .

9.

- (a) Show that the set $\{\mathbf{x}_1, \mathbf{x}_2\}$ generates $\mathbb{R}^2$ for each of the parts in Exercise 6 of this section.
- (b) Show that $\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3\}$ generates $\mathbb{R}^2$ where $\mathbf{x}_1 = (1, 1)$, $\mathbf{x}_2 = (3, 4)$, and $\mathbf{x}_3 = (-1, 5)$.
- (c) Create a set of four or more vectors that generates $\mathbb{R}^2$.
- (d) What is the smallest number of vectors needed to generate $\mathbb{R}^2$? $\mathbb{R}^n$?
- (e) Show that the set

$$\{A_1, A_2, A_3, A_4\} = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\}$$

generates $M_{2 \times 2}(\mathbb{R})$

- (f) Show that $\{1, x, x^2, x^3\}$ generates P^3 .

10. Complete [Example 12.3.16](#) by showing that $\{(1, 1), (-1, 1)\}$ generates $\mathbb{R}^2$.

11.

- (a) Prove that $\{(4, 1), (1, 3)\}$ is a basis for $\mathbb{R}^2$ over $\mathbb{R}$.
- (b) Prove that $\{(1, 0), (3, 4)\}$ is a basis for $\mathbb{R}^2$ over $\mathbb{R}$.
- (c) Prove that $\{(1, 0, -1), (2, 1, 1), (1, -3, -1)\}$ is a basis for $\mathbb{R}^3$ over $\mathbb{R}$.

- (d) Prove that the sets in Exercise 9, parts e and f, form bases of the respective vector spaces.

12.

- (a) Determine the coordinates of the points or vectors $(3, 4)$, $(-1, 1)$, and $(1, 1)$ with respect to the basis $\{(1, 1), (-1, 1)\}$ of $\mathbb{R}^2$. Interpret your results geometrically.
- (b) Determine the coordinates of the points or vector $(3, 5, 6)$ with respect to the basis $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$. Explain why this basis is called the standard basis for $\mathbb{R}^3$.

13.

- (a) Let $\mathbf{y}_1 = (1, 3, 5, 9)$, $\mathbf{y}_2 = (5, 7, 6, 3)$, and $c = 2$. Find $\mathbf{y}_1 + \mathbf{y}_2$ and $c\mathbf{y}_1$.
- (b) Let $f_1(x) = 1 + 3x + 5x^2 + 9x^3$, $f_2(x) = 5 + 7x + 6x^2 + 3x^3$ and $c = 2$. Find $f_1(x) + f_2(x)$ and $cf_1(x)$.
- (c) Let $A = \begin{pmatrix} 1 & 3 \\ 5 & 9 \end{pmatrix}$, $B = \begin{pmatrix} 5 & 7 \\ 6 & 3 \end{pmatrix}$, and $c = 2$. Find $A + B$ and cA .
- (d) Are the vector spaces $\mathbb{R}^4$, P^3 and $M_{2 \times 2}(\mathbb{R})$ isomorphic to each other? Discuss with reference to previous parts of this exercise.

12.4 The Diagonalization Process

12.4.1 Eigenvalues and Eigenvectors

We now have the background to understand the main ideas behind the diagonalization process.

Definition 12.4.1 Eigenvalue, Eigenvector. Let A be an $n \times n$ matrix over $\mathbb{R}$. λ is an eigenvalue of A if for some nonzero column vector $\mathbf{x} \in \mathbb{R}^n$ we have $A\mathbf{x} = \lambda\mathbf{x}$. $\mathbf{x}$ is called an **eigenvector** corresponding to the **eigenvalue** λ .

◇

Example 12.4.2 Examples of eigenvalues and eigenvectors. Find the eigenvalues and corresponding eigenvectors of the matrix $A = \begin{pmatrix} 2 & 1 \\ 2 & 3 \end{pmatrix}$.

We want to find nonzero vectors $X = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ and real numbers λ such that

$$\begin{aligned}
 AX = \lambda X &\Leftrightarrow \begin{pmatrix} 2 & 1 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \lambda \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \\
 &\Leftrightarrow \begin{pmatrix} 2 & 1 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} - \lambda \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \\
 &\Leftrightarrow \begin{pmatrix} 2 & 1 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad (12.4.1) \\
 &\Leftrightarrow \left(\begin{pmatrix} 2 & 1 \\ 2 & 3 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right) \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \\
 &\Leftrightarrow \begin{pmatrix} 2-\lambda & 1 \\ 2 & 3-\lambda \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}
 \end{aligned}$$

The last matrix equation will have nonzero solutions if and only if

$$\det \begin{pmatrix} 2-\lambda & 1 \\ 2 & 3-\lambda \end{pmatrix} = 0$$

or $(2-\lambda)(3-\lambda) - 2 = 0$, which simplifies to $\lambda^2 - 5\lambda + 4 = 0$. Therefore, the solutions to this quadratic equation, $\lambda_1 = 1$ and $\lambda_2 = 4$, are the eigenvalues of A . We now have to find eigenvectors associated with each eigenvalue.

Case 1. For $\lambda_1 = 1$, (12.4.1) becomes:

$$\begin{pmatrix} 2-1 & 1 \\ 2 & 3-1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

which reduces to the single equation, $x_1 + x_2 = 0$. From this, $x_1 = -x_2$. This means the solution set of this equation is (in column notation)

$$E_1 = \left\{ \begin{pmatrix} -c \\ c \end{pmatrix} \middle| c \in \mathbb{R} \right\}$$

So any column vector of the form $\begin{pmatrix} -c \\ c \end{pmatrix}$ where c is any nonzero real number is an eigenvector associated with $\lambda_1 = 1$. The reader should verify that, for example,

$$\begin{pmatrix} 2 & 1 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = 1 \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

so that $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$ is an eigenvector associated with eigenvalue 1.

Case 2. For $\lambda_2 = 4$ (12.4.1) becomes:

$$\begin{pmatrix} 2-4 & 1 \\ 2 & 3-4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \begin{pmatrix} -2 & 1 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

which reduces to the single equation $-2x_1 + x_2 = 0$, so that $x_2 = 2x_1$. The solution set of the equation is

$$E_2 = \left\{ \begin{pmatrix} c \\ 2c \end{pmatrix} \middle| c \in \mathbb{R} \right\}$$

Therefore, all eigenvectors of A associated with the eigenvalue $\lambda_2 = 4$ are of the form $\begin{pmatrix} c \\ 2c \end{pmatrix}$, where c can be any nonzero number. $\square$

The following theorems summarize the most important aspects of the previous example.

Theorem 12.4.3 Characterization of Eigenvalues of a Square Matrix. *Let A be any $n \times n$ matrix over $\mathbb{R}$. Then $\lambda \in \mathbb{R}$ is an eigenvalue of A if and only if $\det(A - \lambda I) = 0$.*

The equation $\det(A - \lambda I) = 0$ is called the **characteristic equation**, and the left side of this equation is called the **characteristic polynomial** of A .

Theorem 12.4.4 Linear Independence of Eigenvectors. *Nonzero eigenvectors corresponding to distinct eigenvalues are linearly independent.*

The solution space of $(A - \lambda I)\mathbf{x} = \mathbf{0}$ is called the eigenspace of A corresponding to λ . This terminology is justified by Exercise 2 of this section.

12.4.2 Diagonalization

We now consider the main aim of this section. Given an $n \times n$ (square) matrix A , we would like to transform A into a diagonal matrix D , perform our tasks with the simpler matrix D , and then describe the results in terms of the given matrix A .

Definition 12.4.5 Diagonalizable Matrix. An $n \times n$ matrix A is called diagonalizable if there exists an invertible $n \times n$ matrix P such that $P^{-1}AP$ is a diagonal matrix D . The matrix P is said to diagonalize the matrix A . $\diamond$

Example 12.4.6 Diagonalization of a Matrix. We will now diagonalize the matrix A of [Example 12.4.2](#). We form the matrix P as follows: Let $P^{(1)}$ be the first column of P . Choose for $P^{(1)}$ any eigenvector from E_1 . We may as well choose a simple vector in E_1 so $P^{(1)} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ is our candidate.

Similarly, let $P^{(2)}$ be the second column of P , and choose for $P^{(2)}$ any eigenvector from E_2 . The vector $P^{(2)} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ is a reasonable choice, thus

$$P = \begin{pmatrix} 1 & 1 \\ -1 & 2 \end{pmatrix} \text{ and } P^{-1} = \frac{1}{3} \begin{pmatrix} 2 & -1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} \frac{2}{3} & -\frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} \end{pmatrix}$$

so that

$$P^{-1}AP = \frac{1}{3} \begin{pmatrix} 2 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix}$$

Notice that the elements on the main diagonal of D are the eigenvalues of A , where D_{ii} is the eigenvalue corresponding to the eigenvector $P^{(i)}$. $\square$

Note 12.4.7

- (1) The first step in the diagonalization process is the determination of the eigenvalues. The ordering of the eigenvalues is purely arbitrary. If we designate $\lambda_1 = 4$ and $\lambda_2 = 1$, the columns of P would be interchanged and D would be $\begin{pmatrix} 4 & 0 \\ 0 & 1 \end{pmatrix}$ (see Exercise 3b of this section). Nonetheless, the final outcome of the application to which we are applying the diagonalization process would be the same.
- (2) If A is an $n \times n$ matrix with distinct eigenvalues, then P is also an $n \times n$ matrix whose columns $P^{(1)}, P^{(2)}, \dots, P^{(n)}$ are n linearly independent vectors.

Example 12.4.8 Diagonalization of a 3 by 3 matrix. Diagonalize the matrix

$$A = \begin{pmatrix} 1 & 12 & -18 \\ 0 & -11 & 18 \\ 0 & -6 & 10 \end{pmatrix}.$$

First, we find the eigenvalues of A .

$$\begin{aligned}\det(A - \lambda I) &= \det \begin{pmatrix} 1 - \lambda & 12 & -18 \\ 0 & -\lambda - 11 & 18 \\ 0 & -6 & 10 - \lambda \end{pmatrix} \\ &= (1 - \lambda) \det \begin{pmatrix} -\lambda - 11 & 18 \\ -6 & 10 - \lambda \end{pmatrix} \\ &= (1 - \lambda)((-\lambda - 11)(10 - \lambda) + 108) = (1 - \lambda)(\lambda^2 + \lambda - 2)\end{aligned}$$

Hence, the equation $\det(A - \lambda I)$ becomes

$$(1 - \lambda)(\lambda^2 + \lambda - 2) = -(\lambda - 1)^2(\lambda + 2)$$

Therefore, our eigenvalues for A are $\lambda_1 = -2$ and $\lambda_2 = 1$. We note that we do not have three distinct eigenvalues, but we proceed as in the previous example.

Case 1. For $\lambda_1 = -2$ the equation $(A - \lambda I)\mathbf{x} = \mathbf{0}$ becomes

$$\begin{pmatrix} 3 & 12 & -18 \\ 0 & -9 & 18 \\ 0 & -6 & 12 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

We can row reduce the matrix of coefficients to $\begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{pmatrix}$.

The matrix equation is then equivalent to the equations $x_1 = -2x_3$ and $x_2 = 2x_3$. Therefore, the solution set, or eigenspace, corresponding to $\lambda_1 = -2$ consists of vectors of the form

$$\begin{pmatrix} -2x_3 \\ 2x_3 \\ x_3 \end{pmatrix} = x_3 \begin{pmatrix} -2 \\ 2 \\ 1 \end{pmatrix}$$

Therefore $\begin{pmatrix} -2 \\ 2 \\ 1 \end{pmatrix}$ is an eigenvector corresponding to the eigenvalue $\lambda_1 = -2$, and can be used for our first column of P :

$$P = \begin{pmatrix} -2 & ? & ? \\ 2 & ? & ? \\ 1 & ? & ? \end{pmatrix}$$

Before we continue we make the observation: E_1 is a subspace of $\mathbb{R}^3$ with basis $\{P^{(1)}\}$ and $\dim E_1 = 1$.

Case 2. If $\lambda_2 = 1$, then the equation $(A - \lambda I)\mathbf{x} = \mathbf{0}$ becomes

$$\begin{pmatrix} 0 & 12 & -18 \\ 0 & -12 & 18 \\ 0 & -6 & 9 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

Without the aid of any computer technology, it should be clear that all three equations that correspond to this matrix equation are equivalent to $2x_2 - 3x_3 = 0$, or $x_2 = \frac{3}{2}x_3$. Notice that x_1 can take on any value, so any vector of the form

$$\begin{pmatrix} x_1 \\ \frac{3}{2}x_3 \\ x_3 \end{pmatrix} = x_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + x_3 \begin{pmatrix} 0 \\ \frac{3}{2} \\ 1 \end{pmatrix}$$

will solve the matrix equation.

We note that the solution set contains two independent variables, x_1 and x_3 . Further, note that we cannot express the eigenspace E_2 as a linear combination of a single vector as in Case 1. However, it can be written as

$$E_2 = \left\{ x_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + x_3 \begin{pmatrix} 0 \\ \frac{3}{2} \\ 1 \end{pmatrix} \mid x_1, x_3 \in \mathbb{R} \right\}.$$

We can replace any vector in a basis with a nonzero multiple of that vector. Simply for aesthetic reasons, we will multiply the second vector that generates E_2 by 2. Therefore, the eigenspace E_2 is a subspace of $\mathbb{R}^3$ with basis

$$\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 3 \\ 2 \end{pmatrix} \right\} \text{ and so } \dim E_2 = 2.$$

What this means with respect to the diagonalization process is that $\lambda_2 = 1$ gives us both Column 2 and Column 3 the diagonalizing matrix. The order is not important so we have

$$P = \begin{pmatrix} -2 & 1 & 0 \\ 2 & 0 & 3 \\ 1 & 0 & 2 \end{pmatrix}$$

The reader can verify (see Exercise 5 of this section) that $P^{-1} = \begin{pmatrix} 0 & 2 & -3 \\ 1 & 4 & -6 \\ 0 & -1 & 2 \end{pmatrix}$ and $P^{-1}AP = \begin{pmatrix} -2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ □

In doing [Example 12.4.8](#), the given 3×3 matrix A produced only two, not three, distinct eigenvalues, yet we were still able to diagonalize A . The reason we were able to do so was because we were able to find three linearly independent eigenvectors. Again, the main idea is to produce a matrix P that does the diagonalizing. If A is an $n \times n$ matrix, P will be an $n \times n$ matrix, and its n columns must be linearly independent eigenvectors. The main question in the study of diagonalizability is “When can it be done?” This is summarized in the following theorem.

Theorem 12.4.9 A condition for diagonalizability. *Let A be an $n \times n$ matrix. Then A is diagonalizable if and only if A has n linearly independent eigenvectors.*

Proof. Outline of a proof: ($\Leftarrow$) Assume that A has linearly independent eigenvectors, $P^{(1)}, P^{(2)}, \dots, P^{(n)}$, with corresponding eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$. We want to prove that A is diagonalizable. Column i of the $n \times n$ matrix AP is $AP^{(i)}$ (see Exercise 7 of this section). Then, since the $P^{(i)}$ is an eigenvector of A associated with the eigenvalue λ_i we have $AP^{(i)} = \lambda_i P^{(i)}$ for $i = 1, 2, \dots, n$. But this means that $AP = PD$, where D is the diagonal matrix with diagonal entries $\lambda_1, \lambda_2, \dots, \lambda_n$. If we multiply both sides of the equation by P^{-1} we get the desired $P^{-1}AP = D$.

($\Rightarrow$) The proof in this direction involves a concept that is not covered in this text (rank of a matrix); so we refer the interested reader to virtually any linear algebra text for a proof. ■

We now give an example of a matrix that is not diagonalizable.

Example 12.4.10 A Matrix that is Not Diagonalizable. Let us attempt

to diagonalize the matrix $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 1 \\ 1 & -1 & 4 \end{pmatrix}$

First, we determine the eigenvalues.

$$\begin{aligned} \det(A - \lambda I) &= \det \begin{pmatrix} 1-\lambda & 0 & 0 \\ 0 & 2-\lambda & 1 \\ 1 & -1 & 4-\lambda \end{pmatrix} \\ &= (1-\lambda) \det \begin{pmatrix} 2-\lambda & 1 \\ -1 & 4-\lambda \end{pmatrix} \\ &= (1-\lambda)((2-\lambda)(4-\lambda) + 1) \\ &= (1-\lambda)(\lambda^2 - 6\lambda + 9) \\ &= (1-\lambda)(\lambda - 3)^2 \end{aligned}$$

Therefore there are two eigenvalues, $\lambda_1 = 1$ and $\lambda_2 = 3$. Since λ_1 is an eigenvalue of degree one, it will have an eigenspace of dimension 1. Since λ_2 is a double root of the characteristic equation, the dimension of its eigenspace must be 2 in order to be able to diagonalize.

Case 1. For $\lambda_1 = 1$, the equation $(A - \lambda I)\mathbf{x} = \mathbf{0}$ becomes

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & -1 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

Row reduction of this system reveals one free variable and eigenspace

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -4x_3 \\ -x_3 \\ x_3 \end{pmatrix} = x_3 \begin{pmatrix} -4 \\ -1 \\ 1 \end{pmatrix}$$

Hence, $\left\{ \begin{pmatrix} -4 \\ -1 \\ 1 \end{pmatrix} \right\}$ is a basis for the eigenspace of $\lambda_1 = 1$.

Case 2. For $\lambda_2 = 3$, the equation $(A - \lambda I)\mathbf{x} = \mathbf{0}$ becomes

$$\begin{pmatrix} -2 & 0 & 0 \\ 0 & -1 & 1 \\ 1 & -1 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

Once again there is only one free variable in the row reduction and so the dimension of the eigenspace will be one:

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ x_3 \\ x_3 \end{pmatrix} = x_3 \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

Hence, $\left\{ \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right\}$ is a basis for the eigenspace of $\lambda_2 = 3$. This means that $\lambda_2 =$

3 produces only one column for P . Since we began with only two eigenvalues, we had hoped that $\lambda_2 = 3$ would produce a vector space of dimension two, or, in matrix terms, two linearly independent columns for P . Since A does not have three linearly independent eigenvectors A cannot be diagonalized. $\square$

12.4.3 SageMath Note - Diagonalization

We demonstrate how diagonalization can be done in Sage. We start by defining the matrix to be diagonalized, and also declare D and P to be variables.

```
var ('D, P')
A = Matrix(QQ, [[4, 1, 0], [1, 5, 1], [0, 1, 4]]);A
```

```
[4 1 0]
[1 5 1]
[0 1 4]
```

We have been working with “right eigenvectors” since the $\mathbf{x}$ in $A\mathbf{x} = \lambda\mathbf{x}$ is a column vector. It’s not so common but still desirable in some situations to consider “left eigenvectors,” so SageMath allows either one. The `right_eigenmatrix` method returns a pair of matrices. The diagonal matrix, D , with eigenvalues and the diagonalizing matrix, P , which is made up of columns that are eigenvectors corresponding to the eigenvectors of D .

```
(D,P)=A.right_eigenmatrix();(D,P)
```

```
([6 0 0]
 [0 4 0]
 [0 0 3],
 [ 1  1  1]
 [ 2  0 -1]
 [ 1 -1  1])
```

We should note here that P is not unique because even if an eigenspace has dimension one, any nonzero vector in that space will serve as an eigenvector. For that reason, the P generated by Sage isn’t necessarily the same as the one computed by any other computer algebra system such as Mathematica. Here we verify the result for our Sage calculation. Recall that an asterisk is used for matrix multiplication in Sage.

```
P.inverse()*A*P
```

```
[6 0 0]
[0 4 0]
[0 0 3]
```

Here is a second matrix to diagonalize.

```
A2=Matrix(QQ,[[8,1,0],[1,5,1],[0,1,7]]);A2
```

```
[8 1 0]
[1 5 1]
[0 1 7]
```

Here we’ve already specified that the underlying system is the rational numbers. Since the eigenvalues are not rational, Sage will revert to approximate number by default. We’ll just pull out the matrix of eigenvectors this time and display rounded entries.

```
P=A2.right_eigenmatrix()[1]
P.numerical_approx(digits=3)
print('-----')
D=(P.inverse()*A2*P);D.numerical_approx(digits=3)
```

```
[ 4.35  0.000  0.000]
[0.000  7.27  0.000]
```

```
[0.000 0.000 8.38]
```

Finally, we examine how Sage reacts to the matrix from [Example 12.4.10](#) that couldn't be diagonalized. Notice that the last column is a zero column, indicating the absence of one needed eigenvector.

```
A3=Matrix(QQ,[[1, 0, 0],[0,2,1],[1,-1,4]])
(D,P)=A3.right_eigenmatrix();(D,P)
```

```
([1 0 0]
 [0 3 0]
 [0 0 3],
 [ 1      0      0]
 [ 1/4    1      0]
 [-1/4    1      0])
```

12.4.4 Exercises

1.

- (a) List three different eigenvectors of $A = \begin{pmatrix} 2 & 1 \\ 2 & 3 \end{pmatrix}$, the matrix of [Example 12.4.2](#), associated with each of the two eigenvalues 1 and 4. Verify your results.
- (b) Choose one of the three eigenvectors corresponding to 1 and one of the three eigenvectors corresponding to 4, and show that the two chosen vectors are linearly independent.

2.

- (a) Verify that E_1 and E_2 in [Example 12.4.2](#) are vector spaces over $\mathbb{R}$. Since they are also subsets of $\mathbb{R}^2$, they are called subvector-spaces, or subspaces for short, of $\mathbb{R}^2$. Since these are subspaces consisting of eigenvectors, they are called eigenspaces.
- (b) Use the definition of dimension in the previous section to find $\dim E_1$ and $\dim E_2$. Note that $\dim E_1 + \dim E_2 = \dim \mathbb{R}^2$. This is not a coincidence.

3.

- (a) Verify that $P^{-1}AP$ is indeed equal to $\begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix}$, as indicated in [Example 12.4.6](#).
- (b) Choose $P^{(1)} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and $P^{(2)} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ and verify that the new value of P satisfies $P^{-1}AP = \begin{pmatrix} 4 & 0 \\ 0 & 1 \end{pmatrix}$.
- (c) Take two different (from the previous part) linearly independent eigenvectors of the matrix A of [Example 12.4.6](#) and verify that $P^{-1}AP$ is a diagonal matrix.

4.

- (a) Let A be the matrix in [Example 12.4.8](#) and $P = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 2 \end{pmatrix}$. Without doing any actual matrix multiplications, determine the

value of $P^{-1}AP$

(b) If you choose the columns of P in the reverse order, what is $P^{-1}AP$?

5. Diagonalize the following, if possible:

$$\begin{array}{lll} \text{(a)} \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix} & \text{(c)} \begin{pmatrix} 3 & 0 \\ 0 & 4 \end{pmatrix} & \text{(e)} \begin{pmatrix} 6 & 0 & 0 \\ 0 & 7 & -4 \\ 9 & 1 & 3 \end{pmatrix} \\ \text{(b)} \begin{pmatrix} -2 & 1 \\ -7 & 6 \end{pmatrix} & \text{(d)} \begin{pmatrix} 1 & -1 & 4 \\ 3 & 2 & -1 \\ 2 & 1 & -1 \end{pmatrix} & \text{(f)} \begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix} \end{array}$$

6. Diagonalize the following, if possible:

$$\begin{array}{lll} \text{(a)} \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} & \text{(c)} \begin{pmatrix} 2 & -1 \\ 1 & 0 \end{pmatrix} & \text{(e)} \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} \\ \text{(b)} \begin{pmatrix} 2 & 1 \\ 4 & 2 \end{pmatrix} & \text{(d)} \begin{pmatrix} 1 & 3 & 6 \\ -3 & -5 & -6 \\ 3 & 3 & 6 \end{pmatrix} & \text{(f)} \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix} \end{array}$$

7. Let A and P be as in [Example 12.4.8](#). Show that the columns of the matrix AP can be found by computing $AP^{(1)}, AP^{(2)}, \dots, AP^{(n)}$.

8. Prove that if P is an $n \times n$ matrix and D is a diagonal matrix with diagonal entries $d_1, d_2, \dots, d_n$, then PD is the matrix obtained from P , by multiplying column i of P by d_i , $i = 1, 2, \dots, n$.

12.5 Some Applications

A large and varied number of applications involve computations of powers of matrices. These applications can be found in science, the social sciences, economics, engineering, and, many other areas where mathematics is used. We will consider a few diverse examples the mathematics behind these applications here.

12.5.1 Diagonalization

We begin by developing a helpful technique for computing A^m , $m > 1$. If A can be diagonalized, then there is a matrix P such that $P^{-1}AP = D$, where D is a diagonal matrix and

$$A^m = PD^mP^{-1} \text{ for all } m \geq 1 \quad (12.5.1)$$

The proof of this identity was an exercise in Section 5.4. The condition that D be a diagonal matrix is not necessary but when it is, the calculation on the right side is particularly easy to perform. Although the formal proof is done by induction, the reason why it is true is easily seen by writing out an

example such as $m = 3$:

$$\begin{aligned}
 A^3 &= (PDP^{-1})^3 \\
 &= (PDP^{-1})(PDP^{-1})(PDP^{-1}) \\
 &= PD(P^{-1}P)D(P^{-1}P)DP^{-1} \quad \text{by associativity of matrix multiplication} \\
 &= PDIDIDP^{-1} \\
 &= PDDDP^{-1} \\
 &= PD^3P^{-1}
 \end{aligned}$$

Example 12.5.1 Application to Recursion: Matrix Computation of the Fibonacci Sequence. Consider the computation of terms of the Fibonacci Sequence. Recall that $F_0 = 1, F_1 = 1$ and $F_k = F_{k-1} + F_{k-2}$ for $k \geq 2$.

In order to formulate the calculation in matrix form, we introduced the “dummy equation” $F_{k-1} = F_{k-1}$ so that now we have two equations

$$\begin{aligned}
 F_k &= F_{k-1} + F_{k-2} \\
 F_{k-1} &= F_{k-1}
 \end{aligned}$$

If $A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$, these two equations can be expressed in matrix form as

$$\begin{aligned}
 \begin{pmatrix} F_k \\ F_{k-1} \end{pmatrix} &= \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} F_{k-1} \\ F_{k-2} \end{pmatrix} \quad \text{if } k \geq 2 \\
 &= A \begin{pmatrix} F_{k-1} \\ F_{k-2} \end{pmatrix} \\
 &= A^2 \begin{pmatrix} F_{k-2} \\ F_{k-3} \end{pmatrix} \quad \text{if } k \geq 3 \\
 &\text{etc.}
 \end{aligned}$$

We can use induction to prove that if $k \geq 2$,

$$\begin{pmatrix} F_k \\ F_{k-1} \end{pmatrix} = A^{k-1} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Next, by diagonalizing A and using the fact that $A^m = PD^mP^{-1}$, we can show that

$$F_k = \frac{1}{\sqrt{5}} \left(\left(\frac{1+\sqrt{5}}{2} \right)^k - \left(\frac{1-\sqrt{5}}{2} \right)^k \right)$$

Some comments on this example:

- (1) An equation of the form $F_k = aF_{k-1} + bF_{k-2}$, where a and b are given constants, is referred to as a linear homogeneous second-order difference equation. The conditions $F_0 = c_0$ and $F_1 = c_1$, where c_1 and c_2 are constants, are called initial conditions. Those of you who are familiar with differential equations may recognize that this language parallels what is used in differential equations. Difference (aka recurrence) equations move forward discretely; that is, in a finite number of positive steps. On the other hand, a differential equation moves continuously; that is, takes an infinite number of infinitesimal steps.
- (2) A recurrence relationship of the form $S_k = aS_{k-1} + b$, where a and b are constants, is called a first-order difference equation. In order to write out

the sequence, we need to know one initial condition. Equations of this type can be solved similarly to the method outlined in the example by introducing the superfluous equation $1 = 0 \cdot F_{k-1} + 1$ to obtain in matrix equation:

$$\begin{pmatrix} F_k \\ 1 \end{pmatrix} = \begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix} \begin{pmatrix} S_{k-1} \\ 1 \end{pmatrix} \Rightarrow \begin{pmatrix} F_k \\ 1 \end{pmatrix} = \begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix}^k \begin{pmatrix} F_0 \\ 1 \end{pmatrix}$$

□

12.5.2 Path Counting

In the next example, we apply the following theorem, which can be proven by induction.

Theorem 12.5.2 Path Counting Theorem. *If A is the adjacency matrix of a graph with vertices $\{v_1, v_2, \dots, v_n\}$, then the entry $(A^k)_{ij}$ is the number of paths of length k from node v_i to node v_j .*

Example 12.5.3 Counting Paths with Diagonalization. Consider the graph in Figure 12.5.4.

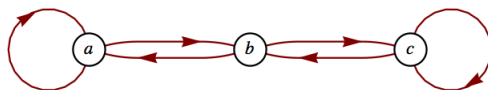


Figure 12.5.4 Counting Numbers of Paths

As we saw in Section 9.2, the adjacency matrix of this graph is $A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$.

Recall that A^k is the adjacency matrix of the relation r^k , where r is the relation $\{(a, a), (a, b), (b, a), (b, c), (c, b), (c, c)\}$ of the above graph. Also recall that in computing A^k , we used Boolean arithmetic. What happens if we use

“regular” arithmetic? If we square A we get $A^2 = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}$

How can we interpret this? We note that $A_{33} = 2$ and that there are two paths of length two from c (the third node) to c . Also, $A_{13} = 1$, and there is one path of length 2 from a to c . The reader should verify these claims by examining the graph.

How do we compute A^k for possibly large values of k ? From the discussion at the beginning of this section, we know that $A^k = PD^kP^{-1}$ if A is diagonalizable. We leave to the reader to show that $\lambda = 1, 2$, and -1 are eigenvalues of A with eigenvectors

$$\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \text{ and } \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$$

Then

$$A^k = P \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2^k & 0 \\ 0 & 0 & (-1)^k \end{pmatrix} P^{-1}$$

where $P = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & -2 \\ -1 & 1 & 1 \end{pmatrix}$ and $P^{-1} = \begin{pmatrix} \frac{1}{2} & 0 & -\frac{1}{2} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{6} & -\frac{1}{3} & \frac{1}{6} \end{pmatrix}$

See [Exercise 12.5.5.5](#) of this section for the completion of this example. $\square$

12.5.3 Matrix Calculus

Example 12.5.5 Matrix Calculus - Exponentials. Those who have studied calculus recall that the Maclaurin series is a useful way of expressing many common functions. For example, $e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$. Indeed, calculators and computers use these series for calculations. Given a polynomial $f(x)$, we defined the matrix-polynomial $f(A)$ for square matrices in Chapter 5. Hence, we are in a position to describe e^A for an $n \times n$ matrix A as a limit of polynomials, the partial sums of the series. Formally, we write

$$e^A = I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \cdots = \sum_{k=0}^{\infty} \frac{A^k}{k!}$$

Again we encounter the need to compute high powers of a matrix. Let A be an $n \times n$ diagonalizable matrix. Then there exists an invertible $n \times n$ matrix P such that $P^{-1}AP = D$, a diagonal matrix, so that

$$\begin{aligned} e^A &= e^{PDP^{-1}} \\ &= \sum_{k=0}^{\infty} \frac{(PDP^{-1})^k}{k!} \\ &= P \left(\sum_{k=0}^{\infty} \frac{D^k}{k!} \right) P^{-1} \end{aligned}$$

The infinite sum in the middle of this final expression can be easily evaluated if D is diagonal. All entries of powers off the diagonal are zero and the i^{th} entry of the diagonal is

$$\left(\sum_{k=0}^{\infty} \frac{D^k}{k!} \right)_{ii} = \sum_{k=0}^{\infty} \frac{D_{ii}^k}{k!} = e^{D_{ii}}$$

For example, if $A = \begin{pmatrix} 2 & 1 \\ 2 & 3 \end{pmatrix}$, the first matrix we diagonalized in Section 12.3, we found that $P = \begin{pmatrix} 1 & 1 \\ -1 & 2 \end{pmatrix}$ and $D = \begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix}$.

Therefore,

$$\begin{aligned} e^A &= \begin{pmatrix} 1 & 1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} e & 0 \\ 0 & e^4 \end{pmatrix} \begin{pmatrix} \frac{2}{3} & -\frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} \end{pmatrix} \\ &= \begin{pmatrix} \frac{2e}{3} + \frac{e^4}{3} & -\frac{e}{3} + \frac{e^4}{3} \\ -\frac{2e}{3} + \frac{2e^4}{3} & \frac{e}{3} + \frac{2e^4}{3} \end{pmatrix} \\ &\approx \begin{pmatrix} 20.0116 & 17.2933 \\ 34.5866 & 37.3049 \end{pmatrix} \end{aligned}$$

$\square$

Remark 12.5.6 Many of the ideas of calculus can be developed using matrices.

For example, if $A(t) = \begin{pmatrix} t^3 & 3t^2 + 8t \\ e^t & 2 \end{pmatrix}$ then $\frac{dA(t)}{dt} = \begin{pmatrix} 3t^2 & 6t + 8 \\ e^t & 0 \end{pmatrix}$

Many of the basic formulas in calculus are true in matrix calculus. For example,

$$\frac{d(A(t) + B(t))}{dt} = \frac{dA(t)}{dt} + \frac{dB(t)}{dt}$$

and if A is a constant matrix,

$$\frac{de^{At}}{dt} = Ae^{At}$$

Matrix calculus can be used to solve systems of differential equations in a similar manner to the procedure used in ordinary differential equations.

12.5.4 SageMath Note - Matrix Exponential

Sage's matrix exponential method is `exp`.

```
A=Matrix(QQ,[[2,1],[2,3]])
A.exp()
```

```
[1/3*e^4 + 2/3*e 1/3*e^4 - 1/3*e]
[2/3*e^4 - 2/3*e 2/3*e^4 + 1/3*e]
```

12.5.5 Exercises

1.

- (a) Write out all the details of [Example 12.5.1](#) to show that the formula for F_k given in the text is correct.
- (b) Use induction to prove the assertion made in the example that

$$\begin{pmatrix} F_k \\ F_{k-1} \end{pmatrix} = A^{k-1} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

2.

- (a) Do [Example 8.3.14](#) using the method outlined in [Example 12.5.1](#). Note that the terminology characteristic equation, characteristic polynomial, and so on, introduced in Chapter 8, comes from the language of matrix algebra,
- (b) What is the significance of [Algorithm 8.3.12](#), part c, with respect to this section?

3. Solve $S(k) = 5S(k-1) + 4$, with $S(0) = 0$, using the method of this section.
4. How many paths are there of length 6 from vertex 1 to vertex 3 in [Figure 12.5.7](#)? How many paths from vertex 2 to vertex 2 of length 6 are there?

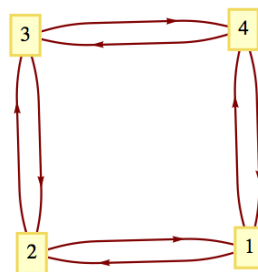


Figure 12.5.7 Graph for exercise 4

Hint. The characteristic polynomial of the adjacency matrix is $\lambda^4 - 4\lambda^2$.

5. Regarding [Example 12.5.3](#),
 - (a) Use matrices to determine the number of paths of length 1 that exist from vertex a to each of the vertices in the given graph. Verify using the graph. Do the same for vertices b and c .
 - (b) Verify all the details provided in the example.
 - (c) Use matrices to determine the number of paths of length 4 there between each pair of nodes in the graph. Verify your results using the graph.
6. Let $A = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$
 - (a) Find e^A
 - (b) Recall that $\sin x = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)!}$ and compute $\sin A$.
 - (c) Formulate a reasonable definition of the natural logarithm of a matrix and compute $\ln A$.
7. We noted in Chapter 5 that since matrix algebra is not commutative under multiplication, certain difficulties arise. Let $A = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 0 \\ 0 & 2 \end{pmatrix}$.
 - (a) Compute e^A , e^B , and e^{A+B} . Compare $e^A e^B$, $e^B e^A$ and e^{A+B} .
 - (b) Show that if $\mathbf{0}$ is the 2×2 zero matrix, then $e^{\mathbf{0}} = I$.
 - (c) Prove that if A and B are two matrices that do commute, then $e^{A+B} = e^A e^B$, thereby proving that e^A and e^B commute.
 - (d) Prove that for any matrix A , $(e^A)^{-1} = e^{-A}$.
8. Another observation for adjacency matrices: For the matrix in [Example 12.5.3](#), note that the sum of the elements in the row corresponding to the node a (that is, the first row) gives the outdegree of a . Similarly, the sum of the elements in any given column gives the indegree of the node corresponding to that column.

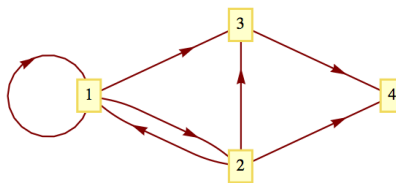


Figure 12.5.8 Graph for exercise 8

- (a) Using the matrix A of Example 12.5.3, find the outdegree and the indegree of each node. Verify by the graph.
- (b) Repeat part (a) for the directed graphs in Figure 12.5.8.

12.6 Linear Equations over the Integers Mod 2

12.6.1 Row reduction mod 2

The methods we have studied for solving systems of equations up to this point can be applied to systems in which all arithmetic is done over other algebraic systems, including the integers modulo 2. The mod 2 case will become particularly useful in our later study of coding theory.

When solving systems of equations with mod 2 arithmetic, the elementary row operations are still fundamental. However, since there is only one nonzero element, 1, you never need to multiply a row by a nonzero constant. One other big difference is that the number of possible solutions is always finite. If you have m linear equations in n unknowns, each unknown can only take on one of two values, 0 or 1. Therefore there are only 2^n possible n -tuples to from which to draw a solution set. Assuming $m \leq n$, you typically (but not always) will have m basic variables after row-reduction and $n - m$ free variable. If this is the case, and any solution exists, there will be 2^{n-m} different solutions.

Let's look at an example, which is converted to matrix form immediately.

$$\begin{array}{cccccc} x_1 + x_2 & & + x_4 & & & = 1 \\ x_1 & & + x_3 & & + x_5 & = 0 \\ & x_2 + x_3 & & & + x_6 & = 0 \end{array}$$

The augmented matrix of the system is

$$\left(\begin{array}{cccccc|c} 1 & 1 & 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 \end{array} \right)$$

The steps in row-reducing this matrix follow. Entries on which we “pivot” are displayed in bold face to more easily identify the basic variables.

$$\begin{aligned} \left(\begin{array}{cccccc|c} 1 & 1 & 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 \end{array} \right) & \xrightarrow{\text{add } R_1 \text{ to } R_2} \left(\begin{array}{cccccc|c} \mathbf{1} & 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 \end{array} \right) \\ & \xrightarrow{\text{add } R_2 \text{ to } R_1} \left(\begin{array}{cccccc|c} \mathbf{1} & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & \mathbf{1} & 1 & 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 \end{array} \right) \\ & \xrightarrow{\text{add } R_2 \text{ to } R_3} \left(\begin{array}{cccccc|c} \mathbf{1} & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & \mathbf{1} & 1 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{array} \right) \end{aligned}$$

Notice that at this point, we cannot pivot on the third row, third column since that entry is zero. Therefore we move over to the next column, making the x_4 basic.

$$\xrightarrow{\text{add } R_3 \text{ to } R_2} \left(\begin{array}{cccccc|c} \mathbf{1} & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & \mathbf{1} & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & \mathbf{1} & 1 & 1 & 1 \end{array} \right)$$

This completes the row reduction and we can now identify the solution set. Keep in mind that since addition is subtraction, terms can be moved to either side of an equals sign without any change in sign. The basic variables are x_1 , x_2 , and x_4 , while the other three variables are free. The general solution of the system is

$$\begin{aligned} x_1 &= x_3 + x_5 \\ x_2 &= x_3 + x_6 \\ x_3 &= x_3 \\ x_4 &= 1 + x_5 + x_6 \\ x_5 &= x_5 \\ x_6 &= x_6 \end{aligned}$$

With three free variables, there are $2^3 = 8$ solutions to this system. For example, one of them is obtained by setting $x_3 = 1$, $x_5 = 1$, and $x_6 = 0$, which produces $(x_1, x_2, x_3, x_4, x_5, x_6) = (0, 1, 1, 0, 1, 0)$.

We can check our row reduction with SageMath:

```
H=Matrix(Integers(2),[[1,1,0,1,0,0,1],
    [1,0,1,0,1,0,0],[0,1,1,0,0,1,0]])
H.echelon_form()
```

```
[1 0 1 0 1 0 0]
[0 1 1 0 0 1 0]
[0 0 0 1 1 1 1]
```

12.6.2 Exercises

In all of the exercises that follow, the systems of equations are over $\mathbb{Z}_2$, and so mod 2 arithmetic should be used in solving them.

1. Solve the following systems, describing the solution sets completely:

$$\begin{aligned} \text{(a)} \quad x_1 + x_2 &= 0 \\ x_1 + x_3 &= 0 \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad x_1 + x_2 &= 0 \\ x_2 + x_3 &= 0 \\ x_3 + x_4 &= 1 \\ x_1 + x_2 + x_3 &= 1 \end{aligned}$$

2. This exercise provides an example in which the number of basic variables is less than the number of equations. The only difference between the two systems below is the right hand sides. You can start with an augmented matrix having two right side columns and do row reduction for both systems at the same time.

$$\begin{aligned} \text{(a)} \quad x_1 + x_2 + x_4 &= 1 \\ x_1 + x_3 + x_4 &= 0 \\ x_2 + x_3 &= 1 \end{aligned}$$

$$\begin{array}{rcl}
 x_1 + x_2 & & + x_4 = 1 \\
 \text{(b)} \quad x_1 & & + x_3 + x_4 = 0 \\
 & x_2 + x_3 & = 0
 \end{array}$$

3. This exercise motivates the concept of a coset in Chapter 15.

- (a) Solve the following system and prove that the solution set is a linear combination of vectors in $\mathbb{Z}_2^5$ and also a subgroup of the group $\mathbb{Z}_2^5$ under coordinatewise mod 2 addition.

$$\begin{array}{rcl}
 x_1 + x_2 & & + x_5 = 0 \\
 x_1 & + x_3 & + x_5 = 0 \\
 x_1 & + x_3 + x_4 & = 0 \\
 & x_2 + x_3 + x_4 & = 0
 \end{array}$$

- (b) Describe the solution set to the following system as it relates to the solution set to the system in the previous part of this exercise.

$$\begin{array}{rcl}
 x_1 + x_2 & & + x_5 = 1 \\
 x_1 & + x_3 & + x_5 = 0 \\
 x_1 & + x_3 + x_4 & = 1 \\
 & x_2 + x_3 + x_4 & = 0
 \end{array}$$